Head-adjunction is not Enough: Deriving Farsi Complex Predicates under Consistent Leftward Movements

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Abstract

In this paper, I address the verb phrase domain in Farsi and the Complex Verb domain, in particular, and highlight empirical data from verb-stranding *v*P ellipsis and from optional verb fronting as pieces of evidence against the traditional SOV analyses and for adopting Kayne (1994) and invariable leftward movements. I argue that head-final verb phrase analyses such as Karimi (2005) and Rasekhi (2018) face inconsistencies with respect to the ordering of auxiliaries and main verbs and will have to rely on ad hoc head movement operations to both the left and right of licensing heads and involving both excorporation and incorporation. The focus of this discussion will be on how such analyses would deal with ellipsis and verb cluster movement in sentences containing a complex predicate and auxiliary verbs, paying special attention to the Future Tense Aux *xâstan* 'want/will'. By proposing an antisymmetric approach to deriving Farsi complex predicates, I argue that verb clusters form phrasal constituents (as opposed to head constituents) in the derivation, and that both ellipsis and verb fronting involve the movement of maximal projections to left-peripheral XP projections. This eliminates the need for excorporation while keeping movement consistently leftward.

Keywords: ellipsis, focus, topic, verb fronting, scrambling, complex predicates, perfect auxiliary, future auxiliary, verbal clusters, antisymmetry, predicate movement, leftward movements

1 Background and introduction

Persian/Farsi has typically been described in the literature as an SOV language where the domain of *v*P/VP is underlying head-final while other categories are generally head-initial (Dabir-Moghaddam 1982, Darzi 1996, Folli, Harley & Karimi 2005, Ganjavi 2007, Ghomeshi 1996, Hashemipour 1989, Karimi 1989, 1994, 2005, Karimi & Smith 2020, Mahootian 1993, Megerdoomian 2012, Rasekhi 2018, among many others). When it comes to simplex verbs, nominal (non-clausal) arguments and adjuncts canonically surface pre-verbally. This is shown in (1) where *pirhan* 'shirt' and *barâ Kimeâ* 'for Kimea' precede *xarid* 'bought'.

(1) *Parviz barâ Kimeâ pirhan* ***xarid***.
Parviz for Kimea shirt bought.3SG
"Parviz bought shirt(s) for Kimea."

Clausal CP arguments of simplex verbs, on the other hand, surface post-verbally, as shown in the position of the CP *ke Kimeâ xune nist* 'that Kimea isn't home' with respect to *goft* 'said/told' in (2).

(2) a. *Parviz be Ali* ***goft*** [*ke Kimeâ xune nist*].
Parviz to Ali said/told.3SG that Kimea home isnt.3SG
"Parviz told Ali that Kimea isn't home."

b. * *Parviz be Ali* [*ke Kimeâ xune nist*] ***goft***.
Parviz to Ali that Kimea home isnt.3SG said/told.3SG

In addition to simplex verbs, Farsi has a wide range of complex predicates, which are verbs formed by a Non-Verbal (NV) predicate (Adj, N, P, Adv,...) and a Light Verb (LV). As (3) shows, non-clausal arguments also precede complex verbs; in this case, the direct object *Kimeâ* precedes the predicate *bidâr kardan* 'awaken'.

(3) *Parviz Kimeâ-ro* ***bidâr kard***.
Parviz Kimea-DOM awake did.3SG
"Parviz awakened Kimea."

Clausal arguments follow complex verbs as they do simplex verbs.[1] In (4), *ke in ketâb-o be-xune* 'to read this book' can only surface after the predicate *ghol dâdan* 'promise'.

(4) a. *Parviz be Ali* ***ghol dâd*** [*ke in ketâb-o be-xune*].
Parviz to Ali promise gave.3SG that this book-DOM SBJV-read.3SG
"Parviz promised Ali to read this book."

b. * *Parviz be Ali* ***ghol*** [*ke in ketâb-o be-xune*] ***dâd***.
Parviz to Ali promise that this book-DOM SBJV-read.3SG gave.3SG
"Parviz promised Ali to read this book."

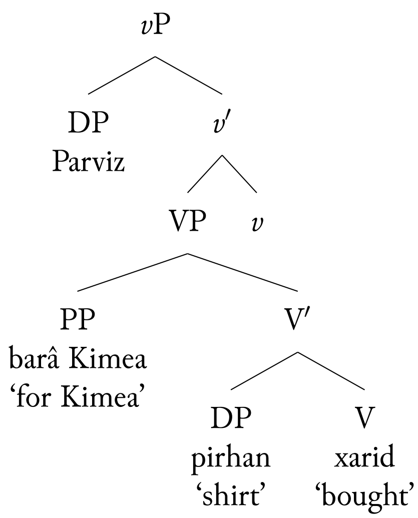
c. * *Parviz be Ali* [*ke in ketâb-o be-xune*] ***ghol dâd***.
Parviz to Ali that this book-DOM SBJV-read.3SG promise gave.3SG

Focusing on complex predicates, Folli, Harley & Karimi (2005) and Karimi (2005) build on the head-final VP structure in (5), and propose that non-verbal predicates head their own small clause projections and introduce the internal argument(s) (similar to V in VP) while light verbs occupy the *v* head and introduce the external argument, as shown in (6)[2].

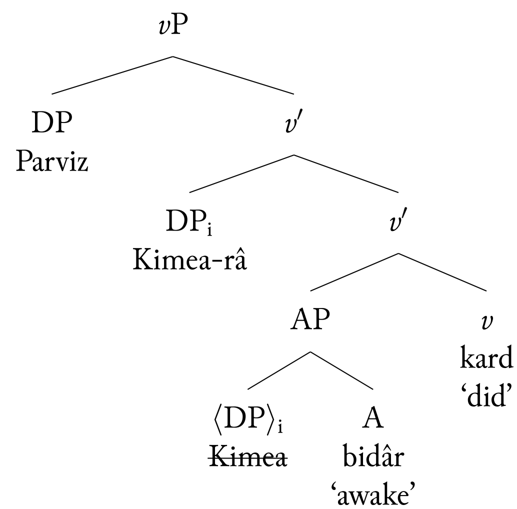
[1] A further elaboration on the post-verbal position of clausal arguments is outside of the scope of this paper. However, see Aghaei 2006, Ahari 2024, Darzi 1996, Farudi 2007, Karimi 2001, 2005, Taleghani 2006, among others, for various accounts of this puzzle.

[2] The movement of the *râ* marked object 'Kimea' follows from Karimi's (2005) analysis that specific direct objects originate as the complement of the predicate and move to the inner [Spec, vP] while non-specific direct objects remain in their base position.

(5) The structure for (1)

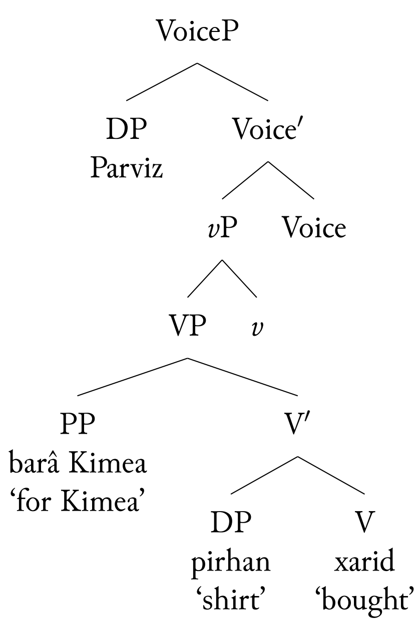


(6) The structure for (3)



In Karimi & Smith (2020), it is assumed that the external argument is instead introduced in VoiceP (Kratzer 1996) and Specifier of *v*P is the position to which specific direct objects move. The structures in (5) and (6) are therefore updated to reflect this in (7) and (8). I will continue my discussion based on Karimi and Smith's (2020) modification, noting that the choice between this and the earlier assumption that the subject originates in *v*P (and the direct object moves to the Spec of the same *v*P) does not bear on the discussion. The crucial observation is the treatment of the components of the complex predicates as heads of their separate projections in either (6) or (8).

(7)



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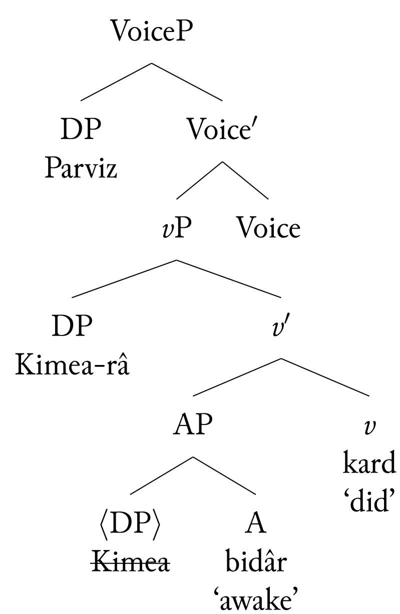


The syntactic independence proposed by Folli, Harley & Karimi (2005) and Karimi (2005) is motivated by empirical evidence that the components of the predicate can be separated in various ways. For instance, the negative prefix *na-* in (9), the habitual/imperfective aspectual marker *mi-* in (10) and inflectional morphemes can intervene between the Non-Verb and the Light Verb of the complex predicate. Lexical units, such as the Future Tense auxiliary *xâstan* 'want', can also break up complex predicates as exemplified in (11).

(9) *Ehsân be man pul* ***gharz*** *na-**dâd**.*
Ehsan to me money loan NEG-gave.3SG
"Ehsan didn't lend me money."

(10) *Ehsân be man pul* ***gharz*** *ne-mi-**dahad**.*
Ehsan to me money loan NEG-IPFV-give.3SG
"Ehsan does not lend me money."

(11) *Ehsân be man pul* ***gharz*** *xâhad* ***dâd**.*
Ehsan to me money loan will/want.3SG give
"Ehsan will lend me money."

Additionally, Folli, Harley & Karimi (2005: p. 7) show in (12) and (13) that adjectives that modify the NV predicate surface inbetween the NV and the LV. They further argue that the possibility for gapping the NV in (14) and the limited scrambling of a quantified NV element in (15) supports their claim that the components of complex predicates are "separately generated and combined in syntax".

(12) *Kimeâ az ra'is-e edâre* [$_{CV}$ [$_{NV}$ ***da'vat**-e* rasmi] ***kard***].
Kimea of boss-Ezf office invitation-Ezf formal did.3SG
"Kimea extended a formal invitation to the boss of the office."

(13) *Kimeâ barâye in xune* [$_{CV}$ [$_{NV}$ ***chune**-ye* xubi] ***zad***].
Kimea for this house chin-Ezf good hit.3SG
"Kimea performed a good negotiation for this house."

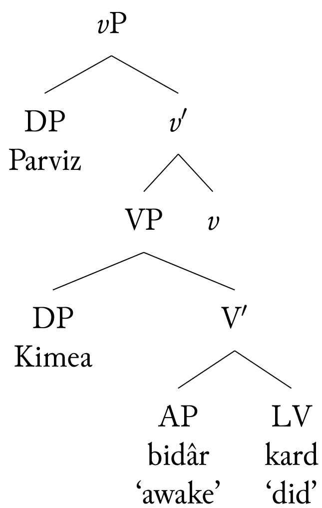
(14) *Kimeâ faghat man-o* ***da'vat*** ***kard**-e,* *to-ro ke* __ *na-**kard**-e.*
Kimea only me-DOM invitation did-PTCP.3SG, you-DOM emph __ NEG-did-PTCP.3SG
"Kimea has only invited me, not you."

(15) *Kimeâ* [*che* ***zamin**-e saxti*]$_i$ *diruz* [$_{CV}$ t_i ***xord***].
Kimea what earth-Ezf hard yesterday collided.3SG
"What a hard fall Kimea had yesterday." Lit. Kimea what a hard earth yesterday collided.

A slightly different approach to the base structure of complex predicates treats the NV and the LV components of the predicate as a syntactic unit (Rasekhi 2018, Shafiei 2015, 2016). Focusing on Rasekhi (2018) in particular, the two elements behave as a constituent that takes internal argument(s) like a V head does.[3] So, extending Rasekhi's structure to (3) yields the following in (16):

[3] Rasekhi does not go into the details of the V′, but she follows Shafiei (2016) in proposing that the [NV+LV] constituent can be targeted for head movement. This suggests that the V′ in this structure ends up being treated as a V node, perhaps following the V′-reanalysis of Larson (1988). Both proposals further claim that the light verb (LV) alone can be targeted for movement to the *v* head, leaving behind the NV in the VP.

(16)



Rasekhi's proposal is motivated by Farsi verb-stranding *v*P ellipsis data such as the following in (17), showing that the verbal element can raise separately or together with the non-verbal item when targeted for head-movement. The sentences (17b-17c) show that in addition to the contrastive PP 'to her son' (contrastive with 'her daughter' in (17a)), either the LV *dâdan* 'give' alone or the NV-LV complex verb *hedye dâdan* 'gift give' together must escape ellipsis (the elided elements are struckthrough).

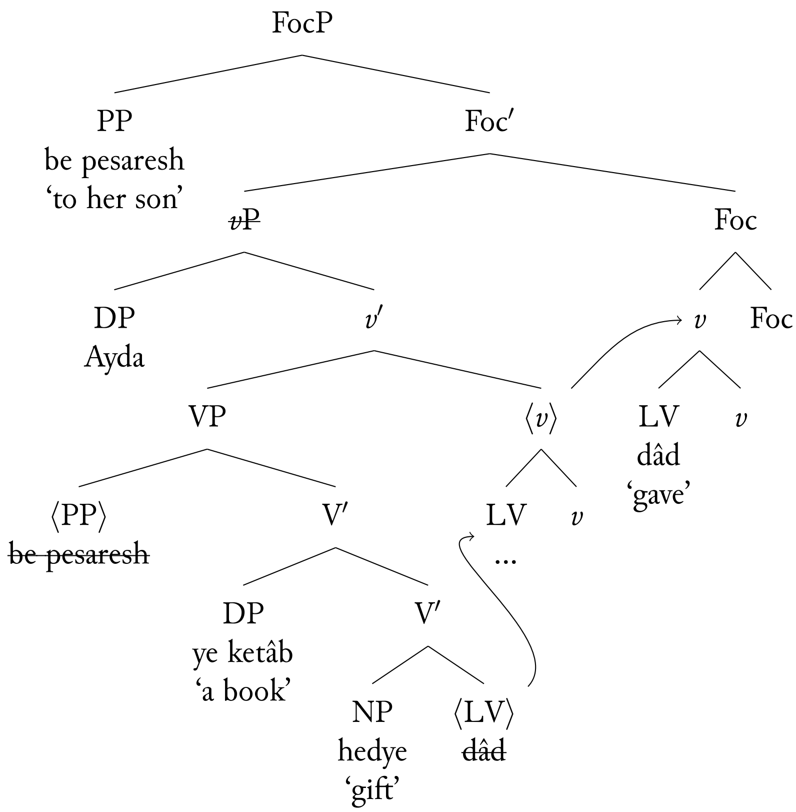
(17) a. *Aydâ be dokhtar-esh ye ketâb* ***hedye*** *na-****dâd****.*
Ayda to daughter-her a book gift NEG.give.3SG
"Ayda didn't gift a book to her daughter."

b. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~* ~~***hedye***~~ ***dâd****.*
but ~~Ayda~~ to son-her ~~a book gift~~ gave.3SG
"But (she) gifted (a book) to her son."

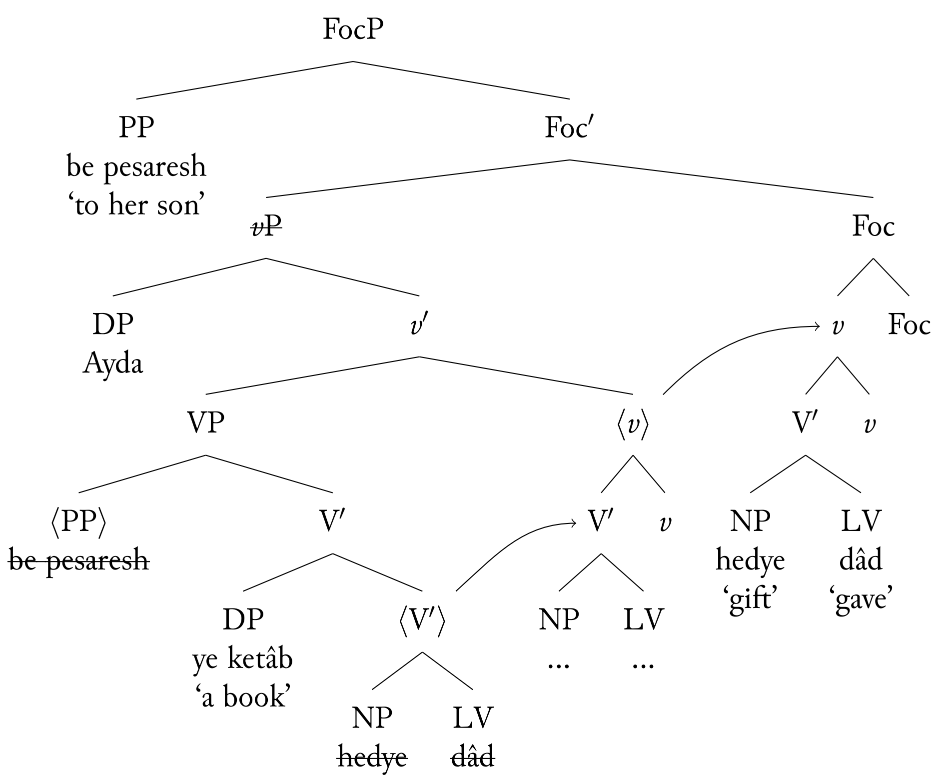
c. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~* ***hedye dâd****.*
but ~~Ayda~~ to son-her ~~a book~~ gift gave.3SG

Rasekhi proposes that in such cases the *v*P is the ellipsis site and gets deleted at the PF level, and the remnants must move out of the *v*P to a TP-internal Foc(us) Phrase. Therefore, the PP raises to the specifier of the FocP and either the LV 'give' in (17b) or the [NV-LV] 'gift give' in (17c) raises to the *v* head and subsequently to the head of FocP. These derivations are given below in (18) and (19), respectively.

(18) Raising the LV alone



(19) Raising the NV-LV complex



2 Some inconsistencies with head-final verb phrase analyses

In this section, I point out some head-adjunction inconsistencies with head-final *v*P analyses that follow from Rasekhi (2018) and Karimi (2005). In section 2.1, I discuss how extending Rasekhi's account of verb-stranding *v*P ellipsis to verbal clusters containing auxiliary verbs such as the Perfect Aux and the Future Tense Aux casts doubts on head-adjunction theories under a head-final verb phrase. Section 2.2 will show that extending Karimi's data from Topic/Focus fronting of verbal elements also falls short in accounting for larger clusters involving these auxiliaries.

2.1 The interaction between Ellipsis and verbal clusters containing auxiliary verbs

While simple cases like the one in (17c) seem to be straightforwardly accounted for by Rasekhi, I will next discuss ellipsis data involving larger verbal clusters that consist of complex predicates and auxiliary verbs (the Perfect and the Future). I will argue that maintaining a head-final *v*P structure has the consequence of imposing (1) an inconsistency in the nature of head-adjunction in terms of whether incorporation is to the left or right of the licensing heads, and (2) a non-uniform head-incorporation approach due to there being a need to resort to an odd excorporation operation out of complex heads.

2.1.1 Ellipsis and the Perfect Auxiliary *budan* 'be'

Building on the data in (17), the addition of the Perfect auxiliary *budan* 'be' in (20) shows both the distribution of this auxiliary with respect to a complex predicate and the interaction between such constructions and the ellipsis phenomena discussed so far. In the non-ellipsis sentence (20a), The Perfect Aux *budan* follows the main predicate *hedye dadan* 'gift give'. Under a standard head-final verb phrase analysis, assuming either Karimi's or Rasekhi's complex predicate configuration, this auxiliary must be assumed to project a higher head-final verbal/aspectual projection selecting for the lexical *v*P, as suggested in (21) and (22).

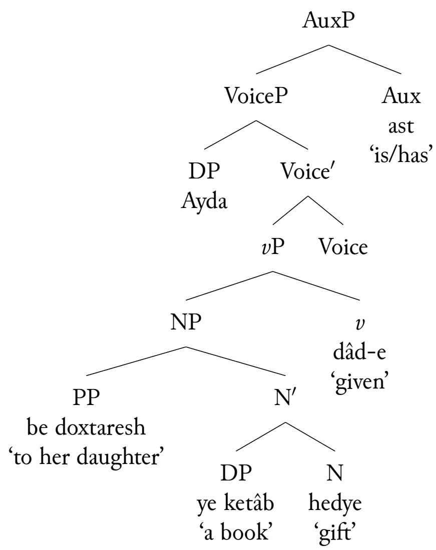
(20) a. *Aydâ be doxtar-esh ye ketâb* ***hedye na-dâd-e*** *ast.*
Ayda to daughter-her a book gift NEG-GIVE-PTCP AUX.PRF.3SG
"Ayda has not gifted a book to her daughter."

b. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~ ~~**hedye**~~ **dâd-e** ast.*
but ~~Ayda~~ to son-her ~~a book gift~~ GIVE-PTCP AUX.PRF.3SG
"But (she) has gifted (a book) to her son."

c. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~ **hedye dâd-e** ast.*
but ~~Ayda~~ to son-her ~~a book~~ gift GIVE-PTCP AUX.PRF.3SG

d. * *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~ ~~**hedye dâd-e**~~ ast.*
but ~~Ayda~~ to son-her ~~a book gift GIVE-PTCP~~ AUX.PRF.3SG

(21)



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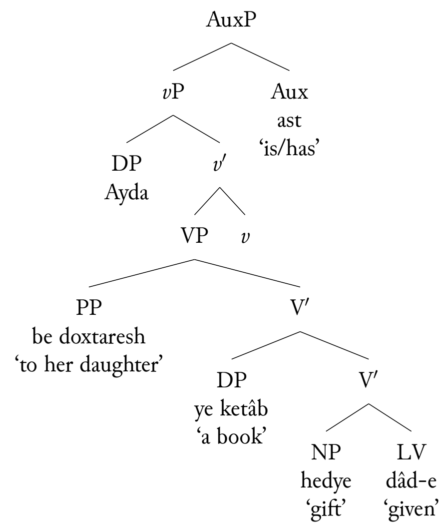
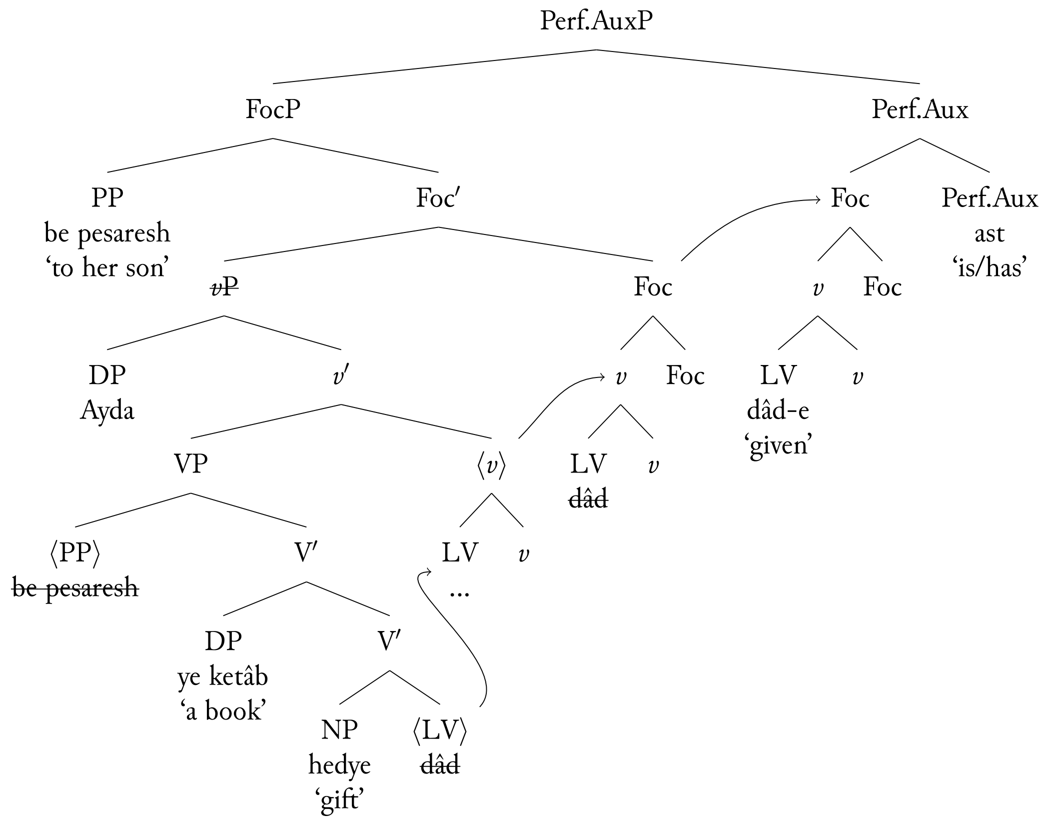


It can further be stipulated that (at least) the verbal component of the complex predicate adjoins to the Perfect auxiliary "overly" (Farudi 2013), given the attachment of the negative prefix *na* 'not' to the participle LV in (20a) and its inability to intervene between the LV and the Perfect auxiliary in (23).

(23) * *Aydâ be doxtar-esh ye ketâb* ***hedye dâd-e*** *na-ast.*
Ayda to daughter-her a book gift GIVE-PTCP NEG-AUX.PRF.3SG
"Ayda has not gifted a book to her daughter."

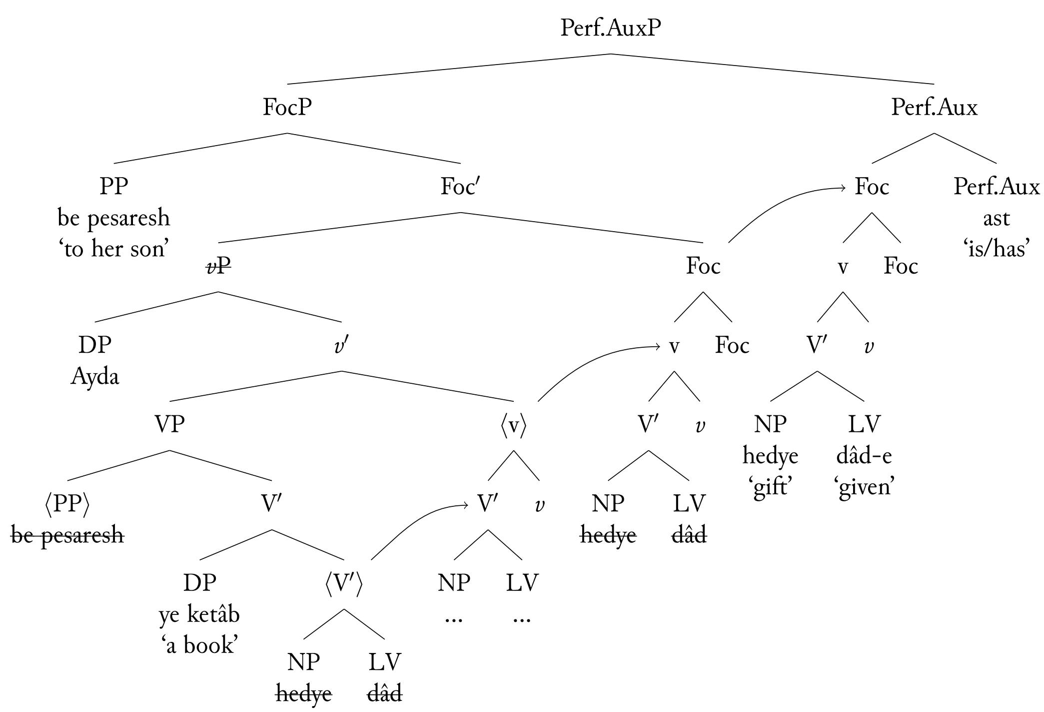
Assuming Rasekhi's complex predicate structure, adjunction to the Perfect auxiliary would therefore involve the movement of either the LV or the NV-LV to the left of the Aux head since both options are proposed to be possible for adjunction to *v* and successively to higher heads. Crucially, incorporation into the Perfect Aux head appears to be possible only to the left of the auxiliary. When paired with ellipsis, (20b) and (20c) show that either the participle light verbal component *dâd-e* 'given' alone or the entire complex predicate *hedye dâde* 'gift given' must be stranded and escape ellipsis. Cases such as (20d), where both components of the complex predicate are deleted and the auxiliary remains, are therefore ungramamtical. Assuming Rasekhi (2018), (20b) would be derived through the adjunction of the light verb to *v* and successively to the Foc head, leaving the non-verbal element in the ellipsis site. Presumably the Foc head to which the LV incorporates further left-adjoins to the Perfect Aux above the FocP, based on the requirement stipulated above for the movement of the verb to this auxiliary. This is shown in (24).

(24) Raising the LV alone



The possibility for the NV element to escape ellipsis alongside the light verb and the aux in (20c) would further be analyzed as the movement of the entire complex predicate constituent to the same higher heads, as shown in (25).

(25) Raising the NV-LV complex



2.1.2 The ellipsis issue with the Future Auxiliary verb *xâstan* 'want'

As observed in the literature (Darzi (1996) and Taleghani (2006); among others), the Future Tense auxiliary verb *xâstan* 'want' differs from the Perfect auxiliary in its distribution with respect to the main predicate.[4] When the Future Aux interacts with a simplex verb in (26a), it does not surface as the final element in the sentence (as evident in the ungrammaticality of (26b)) and must precede the main verb *dâd* 'give'. In such constructions, the auxiliary receives the negation and agreement inflection, and the main verb appears in its (short)infinitive form.

(26) a. *Aydâ be doxtar-esh ye ketâb xâhad* ***dâd***.
Ayda to daughter-her a book AUX.FUT.3SG give
"Ayda will give a book to her daughter."

b. * *Aydâ be doxtar-esh ye ketâb* ***dâd*** *xâhad.*
Ayda to daughter-her a book give AUX.FUT.3SG

In complex predicate constructions, the Future Aux separates the components of the predicate and surfaces penultimately. As (27a) shows, the inflected auxiliary *xâhad* intervenes between the NV *hedye* 'gift' and the infinitive *dâd* 'give' , and it can neither precede the entire complex verb in (27b) nor follow it in (27c).

(27) a. *Aydâ be doxtar-esh ye ketâb* ***hedye*** *xâhad* ***dâd***.
Ayda to daughter-her a book gift AUX.FUT.3SG give
"Ayda will gift a book to her daughter."

b. * *Aydâ be doxtar-esh ye ketâb xâhad* ***hedye dâd***.
Ayda to daughter-her a book AUX.FUT.3SG gift give

c. * *Aydâ be doxtar-esh ye ketâb* ***hedye dâd*** *xâhad.*
Ayda to daughter-her a book gift give AUX.FUT.3SG

According to a head-final *v*P structure, Karimi (2005) and Rasekhi (2018) would need to assume that unlike the Perfect, the Future Aux attracts either the simplex verb in (26a) or light verbal item in (27a) to its right and never to its left. Focusing on the complex predicate cases, the suggested derivations are given in (28) and (29) for Karimi and Rasekhi, respectively.

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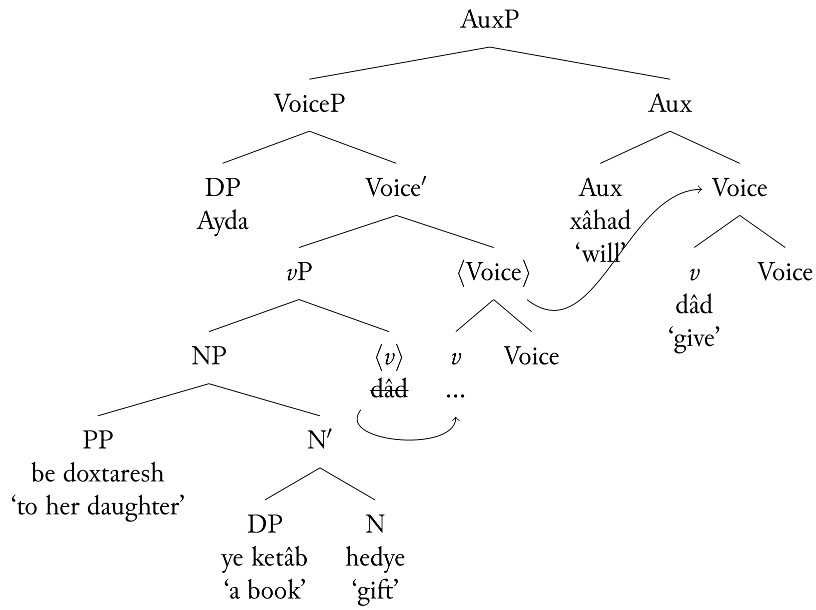
[4]The auxiliary *xâstan* 'want' as used in example (26) and throughout the paper does not function the same way as its control counterpart. Therefore the sentence in (26) cannot also mean 'Ayda wants/wanted to give her daughter a book'. When used as a control verb in the following example, the argument of 'want' is a subjunctive clause as indicated by the subjunctive morpheme *be-* on the embedded verb. In such constructions, both the embedded verb and the matrix verb are inflected for tense and agreement.

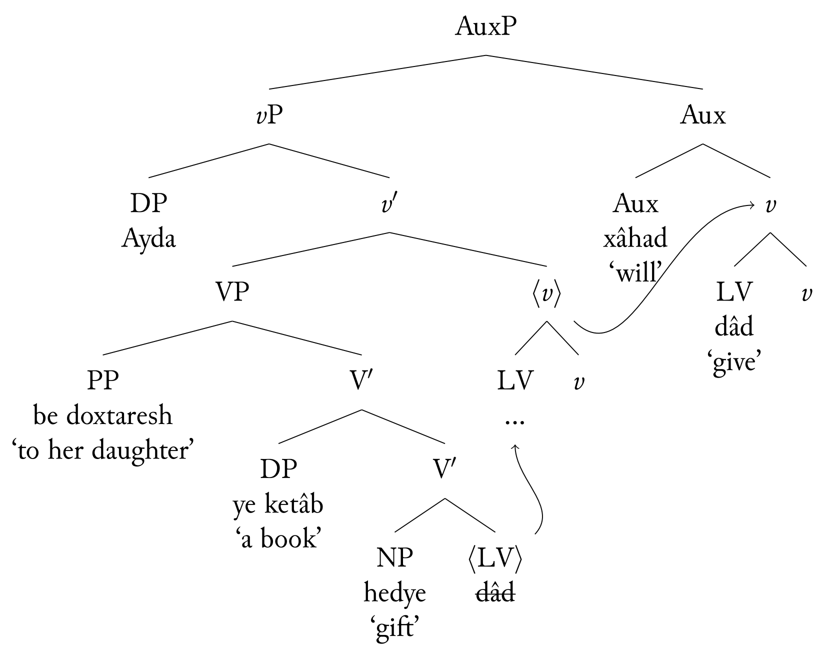
(1) *Aydâ mi-xâhad ke be doxtar-esh ye ketâb be-dahad.*
Ayda wants.3SG that to daughter-her a book SBJV-give.3SG
"Ayda wants to give her daughter a book."

As (26) shows, however, the Future Aux 'want' is the only verbal item inflected for subject agreement and the main verb is "in the form of bare infinitive" (Taleghani 2006).

(28)

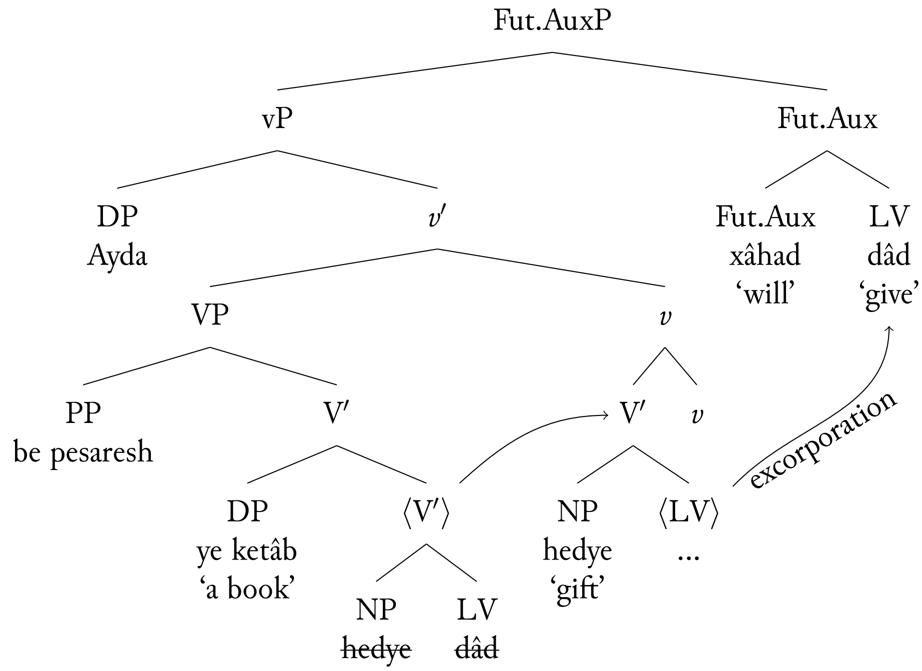


(29)



Therefore, either analysis faces the inconsistency issue of needing to postulate either (a) a head directionality parameter variation for the Future Aux versus the Perfect Aux or (b) adjunction to both the left and the right of licensing heads depending on the auxiliary verb and it not being clear what this asymmetry stems from. Additionally, if Rasekhi's (2018) complex predicate structure and her proposal that the [NV-LV] cluster can move together as a verbal constituent and adjoin to higher heads is on the right track, it remains to be explained as to why movement of this sort should be unavailable for adjunction to the Future Aux and as to what prevents the infelicitous sequences *[Non-Verb Light-Verb Fut.Aux] in (27b) and *[Fut.Aux Non-Verb Light-Verb] in (27c) from surfacing. This shows that the NV and the LV components of a complex predicate cannot consistently incorporate into higher heads as a unit; they appear to be able to incoporate into the *v* head together, but the derivation can only proceed if the LV excorporates out of the complex *v* head and right-adjoins to the Future Aux, as suggested in (30).

(30)



The inability for the entire complex predicate constituent to successively incorporate is further highlighted in ellipsis cases involving the Future Aux. In (31), the verb movement requirement in the ellipsis clause must be captured by either the Aux+LV *xâhad dad* 'will give' together in (31b) or the whole cluster NV+Aux+LV *hedye xâhad dad* 'gift will give' in (31c) escaping deletion.

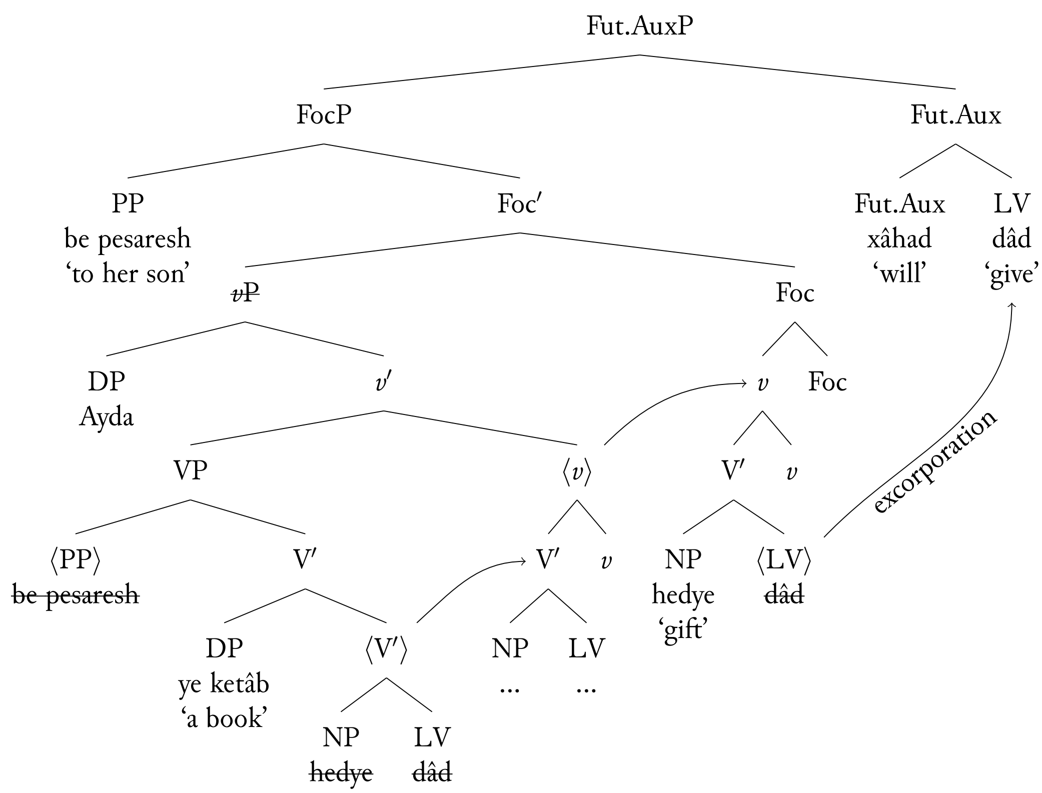
(31) a. *Aydâ be dokhtar-esh ye ketâb* ***hedye*** *na-xâhad* ***dâd****.*
Ayda to daughter-her a book gift NEG-AUX.FUT.3SG give
"Ayda will not gift a book to her daughter."

b. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~* ~~***hedye***~~ *xâhad* ***dâd****.*
but ~~Ayda~~ to son-her ~~a book gift~~ AUX.FUT.3SG give
"But (she) will gift (a book) to her son."

c. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~* ***hedye*** *xâhad* ***dâd****.*
but ~~Ayda~~ to son-her ~~a book~~ gift AUX.FUT.3SG give

To derive (31c), the [NV LV] head *hedye dâd* would have to first incorporate into the *v* head. The complex *v* head should be able to further raise to the Foc head. To account for the position of the Aux in the only possible NV-Aux-LV order, again the LV would have to excorporate out of the complex Foc head to right-adjoin to the Aux as suggested in (32).

(32)



This is not desirable as the derivation would have to impose an ad hoc excorporation requirement inconsistent with all other head movements as incorporation. In addition to excorporation generally being deemed unavailable, the excorporations needed to derive the Future auxiliary data, with or without ellipsis, are not permitted by the head-adjunction theory of excorporation as is generally understood in Roberts (1991) or Koopman (1995). These theories would only allow either the incorporee (the *v* head that adjoins to the Foc head) or the host (the Foc head that *v* adjoins to) to excorporate. Therefore, additional assumptions are needed to account for (30) and (32), where LV targeted for excorporation is neither the incorporee nor the host and must move out of too complex of a head. Alternatively, a head movement analysis that assumes that heads, like phrases, move to the specifier position of probing heads and can subsequently form a complex head with those probing heads via an m-merger operation (Matushansky 2006) would derive excorporation by positing that head movement may take place without the follow-up m-merger. Under such assumptions, the type of excorporation movements needed for Farsi would be similarly ruled out because again only the probing head or the moved head, but not elements internal to them, should be permitted to move.

Given the discussion in this section, two issues stand out so far:

(33) a. There is no consistency to the nature of head-adjunction in terms of whether incorporation is to the left or to the right of licensing heads.
b. There is no uniform head-incorporation approach - excorporation out of complex heads + additional assumptions must also be resorted to.

I will show in the following section that a head-final *v*P analysis is also insufficient in explaining the extended data from the leftward movement of the verbal clusters. This will add to the list of issues in (33).

2.2 Topic/Focus fronting and optional verb raising

2.2.1 Karimi's V-movement account

Karimi (1989, 1994) presents cases of verb preposing in Farsi where the verb may optionality move to the sentence initial position as motivated by discourse functions "contrast", "emphatic", and "interrogative" (Karimi 1989: p. 159). The emphatic interpretation is exemplified in (34) and (35), where the verbs *xaridan* 'buy' and *dâdan* 'give' have fronted, respectively.

(34) a. *man pirhan-o* ***xarid-am***.
I shirt-râ bought-I
"I bought the shirt."

b. ***xarid-am*** *man pirhan-o*. = V S O
bought-I I shirt-râ
"I DID buy the shirt." (Karimi 1989: p. 157)

(35) a. *man be Sâsân pirhan-o* ***dâd-am***.
I to Sasan shirt-râ gave-I
"I gave the shirt to Sasan."

b. ***dâd-am*** *man be Sâsân pirhan-o*. = V S PP O
gave-I I to Sasan shirt-râ
"I DID give the shirt to Sasan." (Karimi 1989: p. 157)

Karimi (2005) observes that the verb may optionally move as enabled by the scrambling of a contrastive nominal element to the left. In (36), the simplex verb *didan* 'see' can optionally front following the scrambling of the direct object 'Kimea'. In (37), the same possibility exists for the verb *dâdan* 'give' following the scrambling of the subject 'Kimea'.

(36) *KIMEA-ro*$_i$ ***didam***$_j$ ⟨*Kimeâ-ro*⟩$_i$ *tu xiâbun* ⟨***didam***⟩$_j$
Kimea-DOM saw.1SG in street
"It was Kimea that I saw in the street." (Karimi 2005: p. 160)

(37) *KIMEA*$_i$ ***dâd***$_j$ ⟨*Kimeâ*⟩$_i$ *ketâb-ro be dust-esh* ⟨***dâd***⟩$_j$
K gave.3SG book-DOM to friend-her
"It was Kimea who gave the book to her friend." (Karimi 2005: p. 160)

Karimi proposes that such cases involve the head movement of the verb to a higher Focus position; in these examples, the subject DPs have moved to the Spec of FocP and the verb has adjoined to Foc. The same sort of verb movement is triggered in topicalization as well, such as in (38), for which Karimi assumes direct object movement to the Spec of TP and verb movement to T.

(38) *un film-ro*$_i$ ***didan***$_j$ *unâ* ⟨*un film-o*⟩$_i$ ⟨*didan*⟩$_j$.
that movie-DOM saw.3SG they
"As for that movie, they saw it." (Karimi 2005: p. 159)

It is shown in Karimi (2001) that verb preposing may move other elements together with the verb. The complex DP data in (39) containing a relative clause shows that the movement of the verb *dâdan* 'give' to the Foc head (whose specifier hosts the focalized direct object 'that book') brings along the indirect object PP *be Kimeâ* 'to Kimea'.

(39) a. *Man* [$_{\text{DP}}$ [*un ketâb-i*]*-ro* [$_{\text{CP}}$ *ke Sepide diruz xar-id*]] <u>*be*</u> <u>*Kimeâ*</u>
I that book-REL-DOM that Sepide yesterday bought.3SG to Kimea
dâd-am.
gave.1SG

"I gave that book to Kimea that Sepide bought yesterday."

b. *Man* [$_{\text{FocP}}$ [*un KETÂB-i-ro*]$_i$ [$_{\text{Foc}}$ <u>*be*</u> <u>*Kimeâ*</u> ***dâd-am***]$_k$] [$_{\text{DP}}$ ⟨*un ketâb-i-ro*⟩$_i$ [$_{\text{CP}}$
I that book-REL-DOM to Kimea gave.1SG
ke Sepide diruz xar-id]] ⟨<u>*be*</u> <u>*Kimeâ*</u> ***dâd-am***⟩$_k$.
that Sepide yesterday bought.3SG

"I gave THAT BOOK (as opposed to a different book) to Kimea that Sepide bought yesterday."

Additionally, (40) shows that the when the verb *xaridan* 'buy' takes a non-specific direct object *xune* 'house', verb movement to Focus must target both elements in (40b).

(40) a. *Man* [$_{\text{DP}}$ [$_{\text{PP}}$ *az mard-i*] [$_{\text{CP}}$ *ke to be man mo'arefi kard-i*]] <u>*xune*</u>
I of man-REL that you to me introduction did.2SG house
xarid-am.
bought.1SG

"I bought a house from a man that you introduced to me."

b. *Man* [$_{\text{FocP}}$ [*az MARD-i*]$_i$ [$_{\text{Foc}}$ <u>*xune*</u> ***xarid-am***]$_k$] [$_{DP}$ ⟨*az mard-i*⟩$_i$ [$_{\text{CP}}$ *ke to be*
I of man-REL house bought.1SG that you to
man mo'arefi kard-i] ⟨<u>*xune*</u> ***xarid-am***⟩$_k$.
me introduction did.2SG

"I bought a house from a MAN that you introduced to me." (not someone else.)

Not only arguments, but also modifers may move with the verb. As (41) shows, raising the negated verb *xândan* 'read' to the Foc head moves both the verb and the modifier *âmadan* 'intentionally' in (41b).

(41) a. *Man* [$_{\text{DP}}$ [*un ketâb-i*]*-ro* [$_{\text{CP}}$ *ke Âlam neveshte bud*]] <u>*amdan*</u>
I that book-REL-DOM that Alam written was.3SG intentionally
na-xund-am.
NEG-read.1SG

"I intentionally did not read that book that Alam had written."

b. *Man* [$_{\text{FocP}}$ [*un KETÂB-i-ro*]$_i$ [$_{\text{Foc}}$ <u>*amdan*</u> ***na-xund-am***]$_k$] [$_{\text{DP}}$
I that book-REL-DOM intentionally NEG-read.1SG
⟨*un ketâb-i-ro*⟩$_i$ [$_{\text{CP}}$ *ke Âlam neveshte bud*]] ⟨<u>*amdan*</u> ***na-xund-am***⟩$_k$.
that Alam written was.3SG

"I intentionally did not read the book that Alam had written." (As opposed to some other book.)

Maintaining a theory of verb-movement to account for the cases discussed so far,[5] Karimi (2001, 2003) proposes the rule of Syntactic Word Formation (SWF), based on Marantz (1997) and Halle & Marantz (1993), to account for the movement of the indirect object and the verb in (39b), the non-specific direct object and the verb in (40b), and the modifier and the verb in (41b). This rule refers to a syntactic operation in which multiple components—such as a verb and its modifiers or arguments—form complex units that denote special meanings and can be targeted for further movement operations. Karimi (2001, 2003) suggests V′ to be the domain on which Syntactic Word Formation operates in Farsi and "where the Encyclopedic knowledge consisting of the special meaning of the verb and its modifiers is obtained". The V′ domain can therefore adjoin to heads of discourse related Topic and Focus projections. From this point of view, the interaction between verb raising and a complex predicate in the following example suggests that what moves must be an X′ level phrase that minimally contains both the NV and the LV elements *gharz dâdan* 'loan give' to raise, as shown in (42c).

(42) a. *Kimeâ ketâb-ro be dust-esh* ***gharz dâd.***
Kimea book-DOM to friend-her loan gave.3SG
"Kimea loaned the book to her friend."

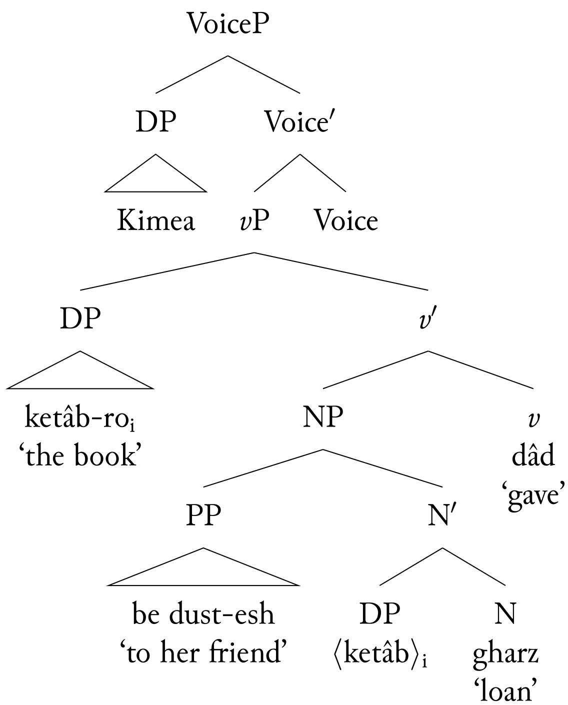
b. * *KIMEA$_i$* ***dâd$_j$*** *⟨Kimeâ⟩$_i$ ketâb-ro be dust-esh* ***gharz*** *⟨**dâd**⟩$_j$*
Kimea gave.3SG book-DOM to friend-her loan
"It was Kimea who loaned the book to her friend."

c. *KIMEA$_i$* ***gharz dâd$_j$*** *⟨Kimeâ⟩$_i$ ketâb-ro be dust-esh ⟨**gharz dâd**⟩$_j$*
Kimea loan gave.3SG book-DOM to friend-her

Since Folli, Harley & Karimi (2005), Karimi (2005), and Karimi & Smith (2020) treat the non-verbal element of a complex verb parallel to a lexical V that introduces the internal arguments, and the light verb in turn occupies the *v* head that selects for the non-verbal phrase, the structure corresponding to (42a) would be as given in (43).

[5]Karimi (2001) argues against analyzing (39b), (40b) and (41b) as rightward CP extraposition due to such movement violating the minimalist principles of (a) being motivated by an uninterpretable feature and (b) applying leftward to a specifier position. The CP would further not be expected to escape the complex DP given the specifier of DP would already be occupied by the determiner and the head noun. Karimi further rules out V′ reanalysis as it would only account for the combination of the verb and the non-specific object in (40) but not for the verb and its modifier in (41).

(43)



Based on this structure, Karimi's (2001, 2003) SWF rule that operates on V′ as its domain would not extend to (43) since V′'s equivalent (N′) contains just the non-verbal element and the copy of the direct object. The SWF domain under the updated complex predicate structure would have to minimally be a *v*′ and it would need be stated as a requirement of verb movement that all verbal elements must prepose together. As the structure stands, however, the first *v*′ also contains the indirect object PP. So, presumably the PP would have to evaculate the NP as well if the *v*′ is what moves in (42c). It is clear that a simple theory of head adjunction does not straightforwardly derive the verb movement data discussed so far. While Karimi utilizes the Syntactic Word Formation rule and specifies a domain for capturing special meanings between the verb and its arguments/modifiers and allowing them to adjoin to heads, this domain tends to expand as large as it gets to in order to contain everything that moves along with the verb. This is specially highlighted by the same Verb-Auxiliary constructions discussed so far, where the Perfect and Future are added to the mix. The following section addresses these cases.

2.2.2 The verb movement issue with the auxiliaries

In sentences containing complex predicates and auxilaries, all the elements in the verbal sequence must move together suggesting that they behave as a unit. In constructions involving the Perfect auxiliary *budan* 'be', such as (44), it is ungrammatical to front the LV *dâde* 'given' alone in (44a), stranding the NV *gharz* 'loan' and the auxiliary. Raising the Aux and the LV (stranding the NV) in (44b) doesn't make it any better. The fronting of the entire NV-LV-Aux cluster in (44c) is the only grammatical option.

(44) a. * *KIMEA$_i$* ***dâd-e$_j$*** *⟨Kimeâ⟩$_i$ ketâb-ro be dust-esh* ***gharz*** *⟨**dâd-e**⟩$_j$*
Kimea give-PTCP book-DOM to friend-her loan
ast.
AUX.PRF.3SG

"It is Kimea who has loaned the book to her friend."

b. * *KIMEA$_i$* ***dâd-e*** *ast$_j$* ⟨*Kimeâ*⟩$_i$ *ketâb-ro be dust-esh* ***gharz***
Kimea give-PTCP AUX.PRF.3SG book-DOM to friend-her loan
⟨***dâd-e*** *ast*⟩$_j$.

c. *KIMEA$_i$* ***gharz dâd-e*** *ast$_j$* ⟨*Kimeâ*⟩$_i$ *ketâb-ro be dust-esh*
K loan give-PTCP AUX.PRF.3SG book-DOM to friend-her
⟨***gharz dâde*** *ast*⟩$_j$

A similar observation is made with respect to sentences containing the Future Aux 'want'. In (45a), it is ungrammatical to move the LV *dâd* alone. Moving just the the Aux and the LV in (45b) is also ruled out. Raising the entire NV-Aux-LV cluster *gharz xâhad dâd* in (45c) is generally the only grammatical option for the speakers.

(45) a. * *KIMEA$_i$* ***dâd$_j$*** ⟨*Kimeâ*⟩$_i$ *ketâb-ro be dust-esh* ***gharz*** *xâhad* ⟨***dâd***⟩$_j$
K give book-DOM to friend-her loan AUX.FUT.3SG
"It is Kimea who will loan the book to her friend."

b. * *KIMEA$_i$ xâhad* ***dâd$_j$*** ⟨*Kimeâ*⟩$_i$ *ketâb-ro be dust-esh* ***gharz*** ⟨*xâhad* ***dâd***⟩$_j$
K AUX.FUT.3SG give book-DOM to friend-her loan

c. *KIMEA$_i$* ***gharz*** *xâhad* ***dâd$_j$*** ⟨*Kimeâ*⟩$_i$ *ketâb-ro be dust-esh*
K loan AUX.FUT.3SG give book-DOM to friend-her
⟨***gharz xâhad dâd***⟩$_j$

Karimi's rule of Syntactic Word Formation is challenged by the auxiliary data because the phrasal domain for the encyclopedic knowledge and obtaining special meanings must also accomodate the auxiliaries. Auxiliary verbs however are not expected to be in that domain because they are not part of the special meanings obtained between the verb and its arguments/modifiers. Therefore, widening this domain to be as large as the intermediate projection including the auxiliary heads is rather too arbitrary and unrestricted. It seems that some (remnant) phrasal movement of the projection containing all the moving elements is neccessary to account for the data, an approach that I will discuss and adopt in section 4. I take this observation as another argument against assuming a head-final *v*P and relying on head-adjunction as the source of movement. Whether one adopts a base structure in line with (a) Folli, Harley & Karimi (2005) and Karimi (2005) or (b) Rasekhi (2018), the extension of these accounts to larger verbal clusters highlights a number of inconsistencies (summarized in (46)) that I consider as grounds for pursuing an alternative approach

(46) a. There is no consistency to the nature of head-adjunction in terms of whether incorporation is to the left or to the right of licensing heads.

b. There is no uniform head-incorporation approach; excorporation out of complex heads + additional assumptions must also be resorted to.

c. There is no straightforward mechanism for deriving leftward verb movement through head-adjunction, without imposing rules for allowing non-head constituents to adjoin to heads.

In the section that follows, I propose an analysis for deriving Farsi complex predicates that assumes a base head-initial structure for all projections following Kayne (1994) and derives the canonical word order through consistent leftward movements.

3 An XP movement analysis for deriving the complex predicate

In this section, I propose that elements that survive deletion in verb-stranding ellipsis data and those that raise in topic/focus fronting data are not complex heads but rather derived XP constituents. More specifically, two types of constituents are formed in the derivation of verbal clusters: (a) a VoiceP/AuxP that contains the light verb (and the auxiliary verb, when it is present), and (b) a PredP that contains the non-verbal element, the light verb, and the auxiliary and forms the complex predicate. This approach unifies ellipsis and fronting under XP movements to the left periphery; while ellipsis allows for either the VoiceP/AuxP or the PredP to escape the ellipsis site, Topic/Focus fronting requires the highest functional projection in the verbal cluster to move. I propose that the derivation of Farsi complex verbs (and all sentences in general) involves uniform leftward movements and left-adjunction, and it conforms to Kayne (1994) Spec-Head-Comp configuration at all steps in the derivation. I assume that the SOV linear order of Farsi is due to a series of feature driven movements that license the arguments as well as the Non-Verbal predicate in the Spec of designated Functional Projections, therefore deriving the preverbal distribution of the predicate's internal arguments (the direct and indirect objects). In section 3.1, I show how this proposal is implemented in detail to derive a complex predicate construction without the auxiliary verbs and how to account for the verb-stranding ellipsis data as a result. Sections 3.2 and 3.3 will subsequently show how this implementation extends for complex predicate constructions involving the Perfect Aux and the Future Aux, and how ellipsis interacts with them.

3.1 Complex predicate derivation without auxiliaries

Starting with (47) (the non-elliptical counterpart to (17c)), I propose that all verb phrases are head-initial and the non-verbal element *hedye* 'gift' acts as the head of a small clause NV Phrase (NVP) which introduces the internal arguments *ye ketâb* 'a book' and *be pesar-esh* 'to her son'. The light verb heads its *v*P projection above NVP. This directly follows Folli, Harley & Karimi (2005) and Karimi (2005) in assuming that the components of the complex verb head their independent projections. In line with Karimi & Smith (2020), the external argument 'Ayda' is further introduced in a VoiceP,[6] head of which attracts the light verb (the *v* head). These steps

[6] There is evidence to argue that the subject merges external to the light verbal *v*P (and not in the specifier of *v*P). One such argument is based on the following example from Megerdoomian (2012) in which the causative is realized on the LV with an additional morpheme *ân*. This suggests that a separate head causativizes the light verb, and introduces the external argument.

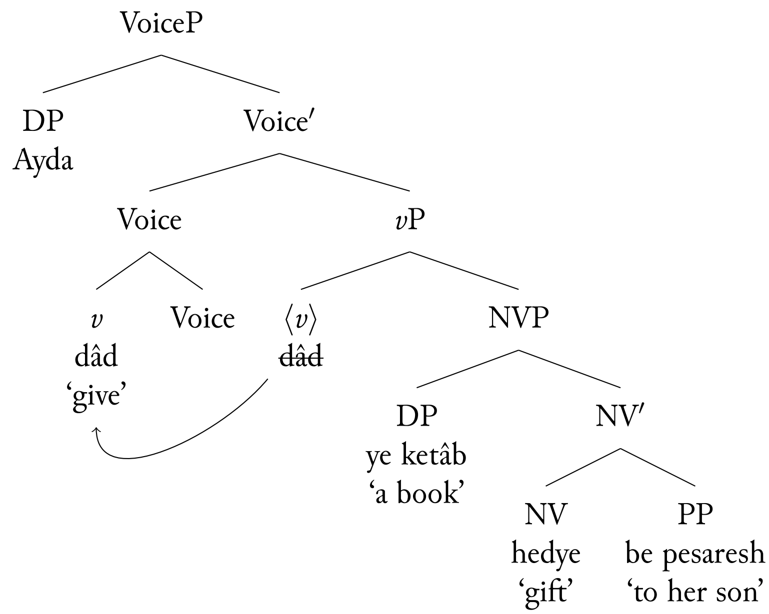
(1) a. *jang* ***be pâyân resid.***
war to end arrived.3SG
"The war came to an end."

b. *soqut-e in rejim jang-râ* ***be pâyân res-ân-d.***
fall-Ezf this regime war-DOM to end arrived-CAUS.3SG
"The fall of this regime brought the war to an end." (Megerdoomian 2012: ex. 67)

are shown in (48).

(47) *Aydâ be pesar-esh ye ketâb* ***hedye dâd.***
Ayda to son-her a book gift gave.3SG
"Ayda gifted a book to her son."

(48)



Taking inspiration from Zwart (1993, 1996, 1997), I assume that a PredP functional projection merges above all other verbal items and below the derived position of the object. Zwart proposes specifier of PredP as a licensing position for small clause predicates and particles in Dutch, and Koster (1999, 2000) assumes that XPs that are introduced by a verb but are not licensed as independent arguments throughout the derivation end up being part of the verb through licensing in spec of PredP. This projection as adopted for Farsi will merge below the licensing position of the object(s) and higher than the position of the (light verb), with its specifier as the landing site for the non-verbal predicate. One of the conditions for licensing this position is that this phrase can only project above the verbal projection that ends up hosting the non-verbal phrase's original licensor (the light verb). This turns out to be the case in (48) since the LV adjoins to Voice. In fact, all complex predicate derivations prepose the non-verbal projection through movement to [Spec,PredP]. Zwart (1997) originally proposed that the Pred head has a strong N feature that must be satisfied locally with the non-verbal predicate in its

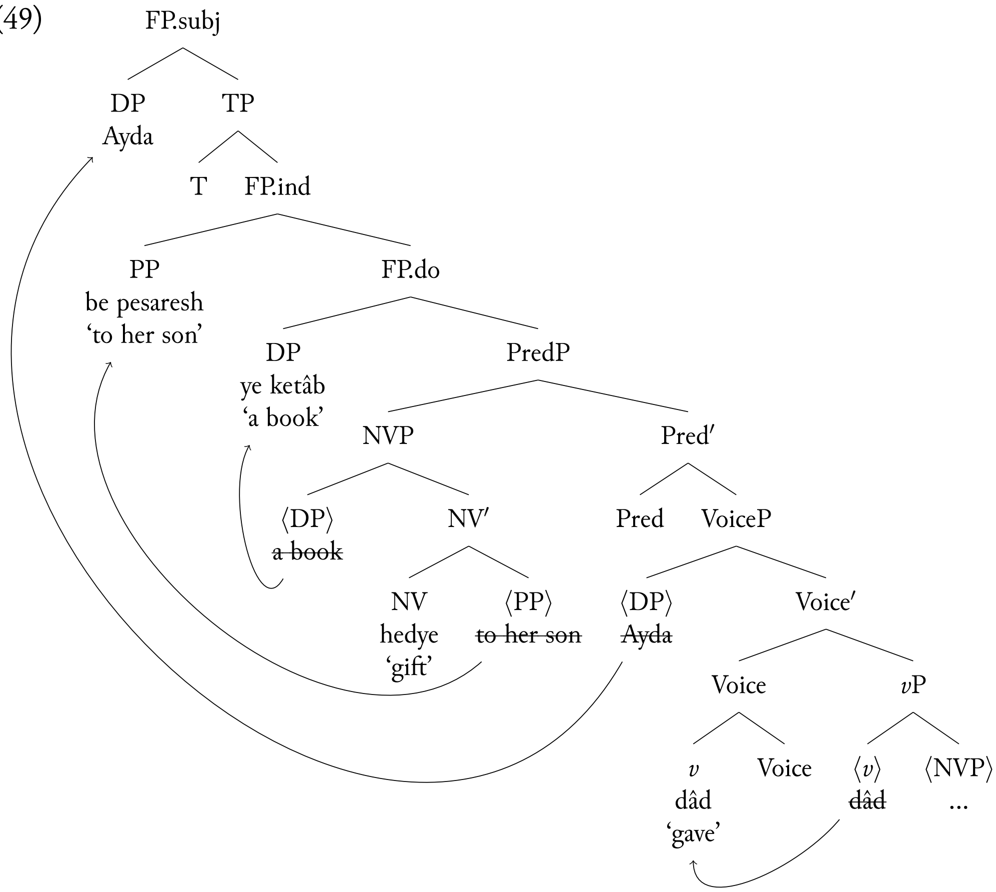
Additionally, (2) below exemplifies a construction in which the passive voice alternative to (2a) is derived with an additional verb *shodan* 'become' in (2b). In this case, the active-passive alternation is not due to an alternation in the light verb *dâd* but rather due to a separate head external to the *v*P.

(2) a. *unâ âpârtemân-o be Ali* ***edʒâre dâdand.***
they apartment-DOM to Ali rent gave.3PL
"They rented the apartment to Ali."

b. *âpârtemân be Ali* ***edʒâre dâd-e*** *shod.*
apartment to Ali rent give-PTCP became.3SG
"The apartment was rented to Ali."

specifier. In my analysis, I take this movement to be a reflex of an Edge feature, leaving out a detailed discussion of whether it must be the head (Chomsky 2001) or the moving element itself (Bošković 2007) hosting the feature. I further assume that the internal arguments inside [Spec,PredP] must move be locally licensed in the specifiers of their Functional Projections (FP.do for the functional projection licensing the direct object and FP.ind for the one licensing the indirect object). The external argument must also raise to its designated licensing functional position FP.subj. These phrases can be alternatively thought of Agr projections for the direct object, the indirect object, and the subject. The full derivation is given in (49) below:[7]

(49)



This derivation is in line with the feature checking approach of Zwart (1993, 1996, 1997), which assumes that a series of licensing operatins apply, triggering the movement of elements from their base merge positions to specifiers of functional licensing projections. In the context of OV languages, this means that such operations take place in the narrow syntax and the licensing is satisfied locally due to the functional heads' EPP features. The movements proposed for deriving the linear order are therefore necessary given the antisymmetric framework of the analysis, and they ultimately follow from a set of universal conditions as well as language-specific choices.I also assume that the licensing movement of the direct object DP takes place independent of whether the object is non-specific as in (47) or specific (*râ*-marked) as in (50) below, in which case it typically precedes the indirect object PP.

[7]Throughout this paper, only the specifiers of FP.do, FP.ind, and FP.subj are represented in the trees.

(50) *Aydâ un ketâb-râ be pesar-esh* ***hedye dâd.***
Ayda that book-DOM to son-her gift gave.3SG
"Ayda gifted that book to her son."

I follow Farudi (2007) and Kahnemuyipour (2004, 2009) in proposing that specifc objects such as in (50) undergo a subsequent movement to a higher position where they receive the *râ* marking.

3.1.1 Ellipsis without the auxiliaries

My account of deriving the verb-stranding ellipsis data discussed in Rasekhi (2018) will essentially differ from Rasekhi in two aspects: (a) I assume that the verb-stranding operation involves maximal projection to specifier movement, rather than head-movement; and (b) I consider the ellipsis site to be the full (FP.subj) clause rather than the *v*P. My analysis is still in line with Rasekhi's proposal that ellipsis is constrained by contrastive focus, and in Farsi this comes in three possible interpretations: contrastive, additive, and corrective. Rasekhi provides evidence that focus plays a role for licensing ellipsis by showing that the verbs across the antecedent and ellipsis clauses can be lexically contrastive and need not be identical. The contrastive interpretation across the two clauses can be shown in the pair in (51), where the verbs *xaridan* 'buy' and *furuxtan* 'sell' are lexically contrastive.

(51) Modified from Rasekhi (2018: p. 71) to remove embedding.

a. *Ali pârsâl yeho un xuna-ro* ***xarid.***
Ali last-year suddently that house.ftdom bought.3SG
"Ali bought that house suddenly last year."

b. *Vali ~~Ali~~ emsâl yeho ~~un xuna-ro~~* ***furuxt.***
but ~~Ali~~ this-year suddently ~~that house.DOM~~ sold.3SG
"But (he) sold (that house) suddenly this year."

In the additive interpretation, the verb 'buy' is identical across the clauses, and the focus particle *ham* 'also' follows the contrastive external argument *Araz* in (52b).

(52) Modified from Rasekhi (2018: p. 72) to remove embedding.

a. *Aydâ âpârtemân* ***xarid.***
Ayda apartment bought.3SG
"Ayda bought an apartment."

b. *Araz ham ~~âpârtemân~~* ***xarid.***
Araz also apartment bought.3SG
"Araz also bought (an apartment)."

In verb-stranding ellipsis cases with a corrective interpretation, verbs can be either contrastive across (53a) and (53b) or identical across (53a) and (53c). If the verbs are lexically identical, however, some other element must be contrastive with its correlate in the antecedent clause.

(53) Modified from Rasekhi (2018: p. 73) to remove embedding.

a. *Aydâ benz-a-ro* ***xarid.***
Ayda Benz-DEF-DOM bought.3SG
"Ayda bought the Mercedez."

b. *na, ~~Aydâ~~ ~~benz-a-ro~~* ***furuxt.***
no ~~Ayda~~ ~~Benz-DEF-DOM~~ sold.3SG
"No, (Ayda) sold (the Mercedez)."

c. *na, Araz ~~benz-a-ro~~* ***xarid.***
no, Araz ~~Benz-DEF-DOM~~ bought.3SG
"no, Araz bought the Mercedez."

The following pairs in (54) and (55) further reveal that a polarity contrast on the verb (affirmative vs. negative) qualifies as the verbs being in a contrastive relationship.

(54) Contrastive interpretation similar to (51).

a. *Ali pârsâl yeho ye xune* ***xarid.***
Ali last-year suddently a house bought.3SG
"Ali suddently bought a house last year."

b. *Vali ~~Ali~~ emsâl ~~yeho~~ ~~ye xune~~ na-****xarid.***
but ~~Ali~~ this-year suddently ~~that house.DOM~~ sold.3SG
"But (he) didn't suddenly buy (a house) this year."

(55) Corrective interpretation similar to (53a) and (53b).

a. *Aydâ benz-a-ro* ***xarid.***
Ayda Benz-DEF-DOM bought.3SG
"Ayda bought the Mercedez."

b. *na, ~~Aydâ~~ ~~benz-a-ro~~ na-****xarid.***
no ~~Ayda~~ ~~Benz-DEF-DOM~~ NEG-bought.3SG
"No, (Ayda) didn't buy (the Mercedez)."

Rasekhi (2018) proposes that there is TP-internal Foc(us) P(hrase) above the *v*P. This Foc head is bundled with a number of features: (a) an [E] feature that licenses the deletion of the *v*P complement at PF; (b) an uninterpretable V feature that is satisfied through the movement of the verb to this Foc head; and (c) an uninterpretable Contrastive Focus feature (ConF) that must be valued through the movement of a contrastively focused argument (contrastive with a correlate in the antecedent clause) to its specifier. To revisit how Raskehi's analysis works, the ellipsis data in (17) is repeated here as (56).

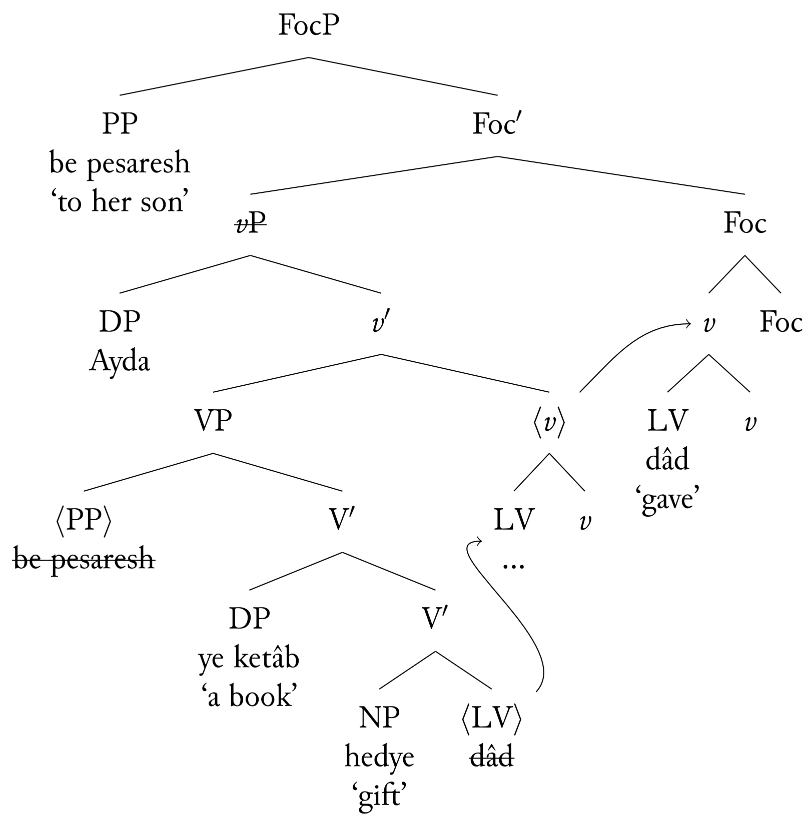
(56) a. *Aydâ be dokhtar-esh ye ketâb* ***hedye*** *na-****dâd.***
Ayda to daughter-her a book gift NEG.give.3SG
"Ayda didn't gift a book to her daughter."

b. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~* ***~~hedye~~ dâd.***
but ~~Ayda~~ to son-her ~~a book gift~~ gave.3SG
"But (she) gifted (a book) to her son."

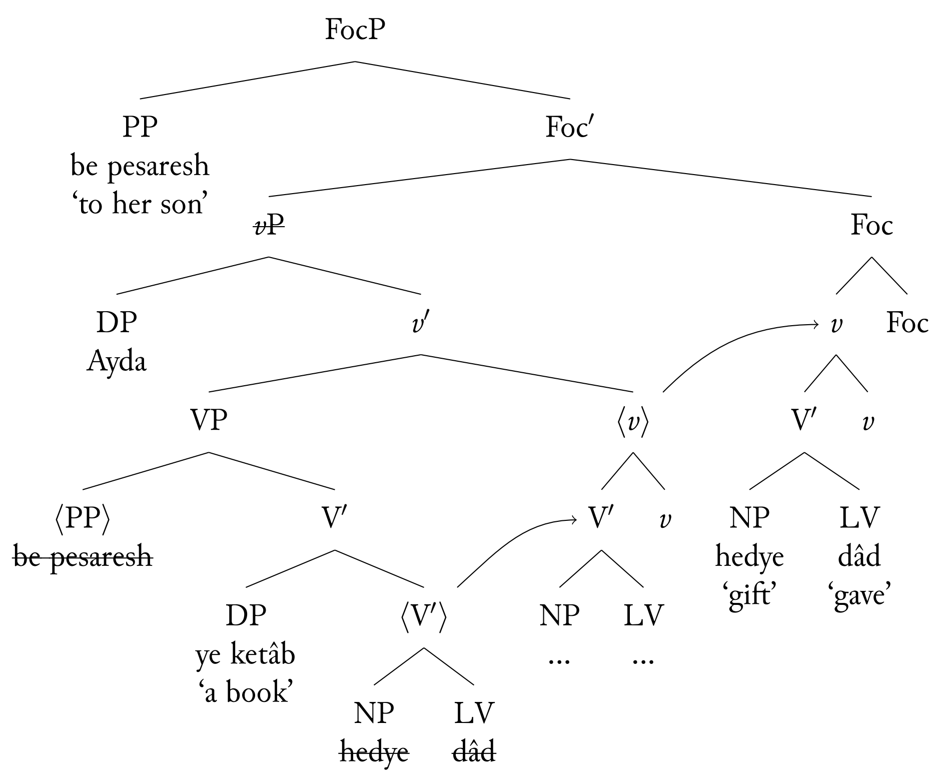
c. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~* ***hedye dâd.***
but ~~Ayda~~ to son-her ~~a book~~ gift gave.3SG

In this case, the PP *be pesaresh* 'to her son' raises to the specifier of the FocP and either the LV 'give' in (56b) or the [NV-LV] 'gift give' in (56c) raises to the *v* head and subsequently to the head of FocP. These derivations are repeated below in (57) and (58), respectively.

(57) Raising the LV alone



(58) Raising the NV-LV complex



My account of the ellipsis data however builds on the alternative derivation laid out in (48). I propose that a functional projection exists in the left periphery of the clause to which a verbal constituent moves from the ellipsis site. This means that whenever an additional contrastive element escapes the ellipsis site, it moves to the specifier position of a separate functional projection than the one utilized for stranding the verb. It was shown in (51)-(53) that there are three contrastive interpretations available to the verb-stranding ellipsis data. It can be further observed that the contrastive focus requirements are not exactly the same for all three interpretations, as summarized in (59):

(59) a. **Contrastive focus interpretation** requires (a) **the verbs to be contrastive** across the ellipsis clause and the antecedent clause, and (b) minimally **one other element to be in a constrastive focus relation** with its correlate.

b. **Additive interpretation** requires (a) **the verbs to be identical** across the clauses, and (b) **a contrastive element followed by the focus particle *ham* 'also'** to escape ellipsis.

c. **Corrective interpretation** can be of two forms:

i. **The verbs are contrastive** across the clauses and **no additional contrastive element** is needed.

ii. **The verbs are identical** and **one other element is contrastive** with its correlate in the antecedent clause.

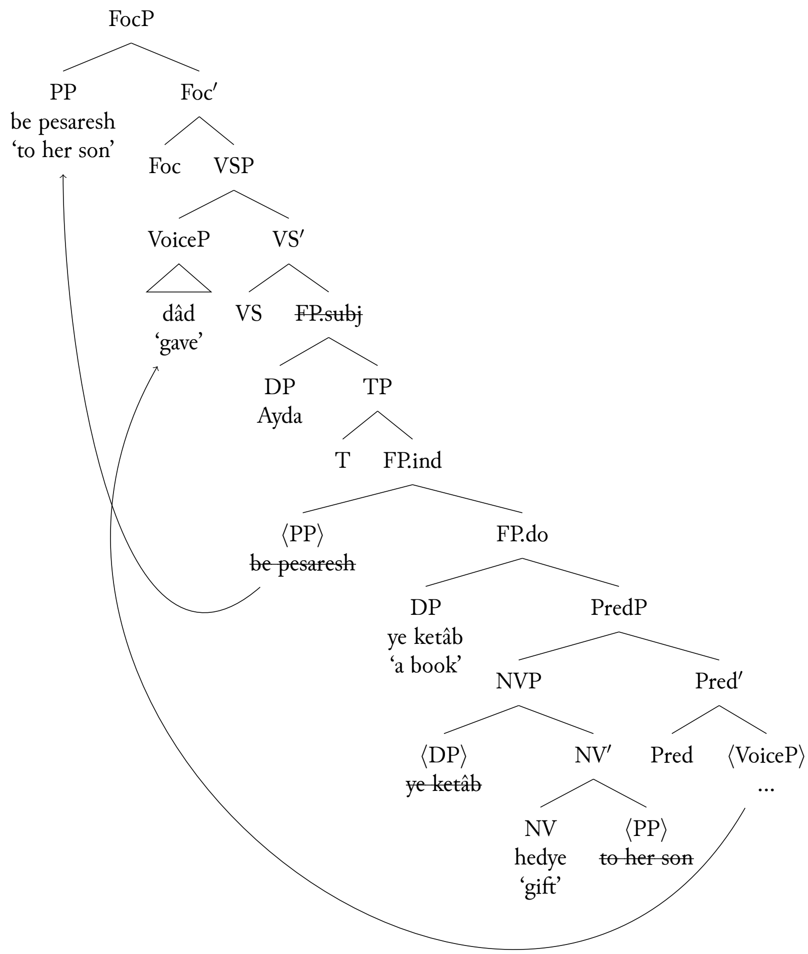
Crucially, while all the ellipsis data show verb stranding, not all constructions have the same identity requirement on the verb and not all constructions require an additional contrastive XP to move to the left periphery (as is the case in (59c-i)). Consequently, I claim that the only projection constant for deriving the different constructions is the one attracting a verbal constituent, which I will label Verb Stranding Phrase (VSP). The VS head of this phrase will carry either a Contrastive Verb (ConV) or Identitcal Verb (IdentV) feature depending on the particular ellipsis construction. In the context of a contrastive construction such as (56), the VS head will necessarily carry the ConV feature. This feature must be locally satisfied by raising either the VoiceP (containing only the LV *dâd*) or the PredP (containing both the NV and the LV *hedye dâd*) to account for (56b) and (56c), respectively. Above this VSP, Farsi requires another projection for raising the contrastive internal argument *be pesaresh*, which I've labled FocP in line with Rasekhi (2018). Therefore, this construction requires both the verbs and minimally an argument to be contrastive.[8] The two verb cluster stranding options in (56b) and (56b) are shown in (60) and (61), respectively.

[8]I'm claiming that the predicates in (56) are contrasted based the observation that the negative-affirmative verb pairs in (54) and (55) behave like lexically contrastive verbs in (51) and (53a-b). Additionally, Rasekhi (2018) suggests that the verbs in the following pair (*na-dâd* 'didn't give' and *dâd* 'gave') are contrastive, which means that the ones in (56) must be as well.

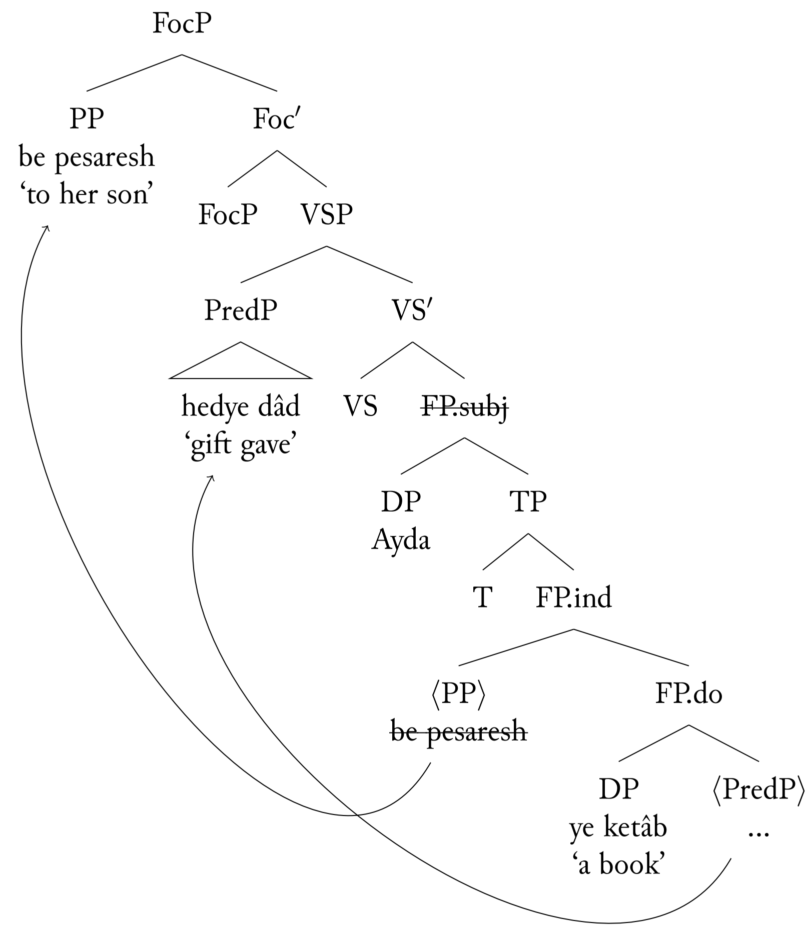
(1) a. *Az in-ke Ayda* ***ketâb-ro*** *be pesar-esh na-dâd ta'ajob na-kard-am.*
from this-that Ayda book-DOM to son-her NEG-gave.3SG surprise NEG-did-1SG
"The fact that Ayda did not give the book to her son did not surprise me."

b. *Vali az in-ke* ***majalla-ro*** *~~be pesar-esh~~ dâd ta'ajob kardam.*
but from this-that magazine-DOM ~~to son-her~~ gave3SG surprise did.1SG
"But the fact that (she) gave the magazine (to her son) surprised me." (Rasekhi 2018: p. 83)

(60) The VoiceP can move to satisfy the VS head's ConV feature



(61) The PredP can move to satisfy the VS head's ConV feature



3.2 Complex predicates with the Perfect Auxiliary

With the addition of auxiliaries into the mix, I follow Zwart (1996) and Barbiers, Cornips & Corrigan (2005) in assuming that all auxiliary verbs enter the syntax according to a hierarchical base structure and syntactic operations further follow a feature checking approach. The Perfect auxiliary *budan* in (62) is one such auxiliary and it licenses participles, as evident from the participle marking '-e' on the light verb.

(62) *Aydâ be pesar-esh ye ketâb* ***hedye dâd-e*** *ast.*
Ayda to son-her a book gift GIVE-PTCP AUX.PRF.3SG
"Ayda has gifted a book to her son."

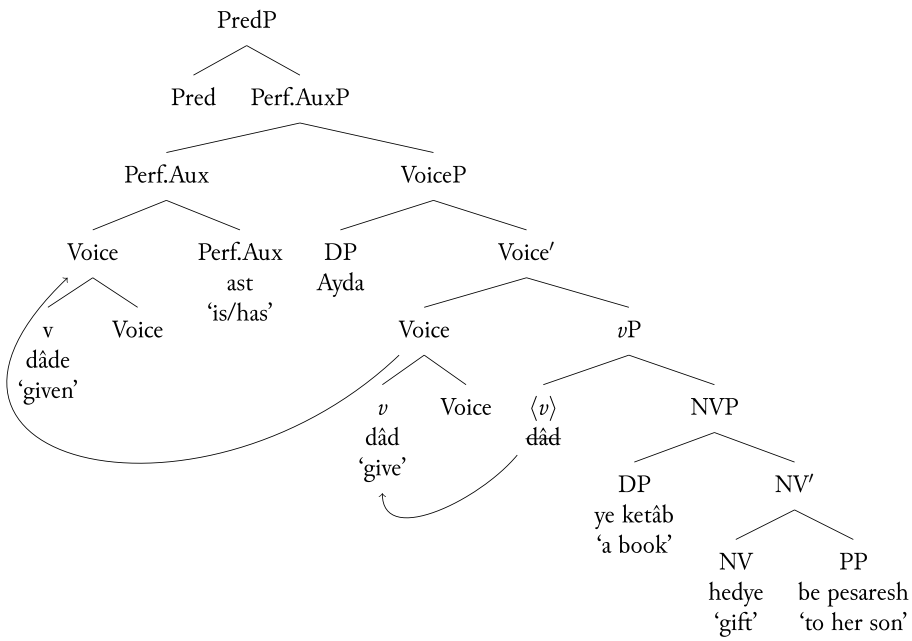
I propose that the participle LV adjoins to the Perfect Aux head resulting in a complex head, and this is the only Auxiliary verb that requires the head movement of the participle it selects for. This explains the observation pointed out above regarding the placement of the negative prefix in such constructions. As (63) shows, the Neg- attaches to the left of the sequence [Participle-Perf.Aux] and never intervenes between the two heads. This is expected if the participle and the Perfect aux form a complex head, and negation is always realized on the highest verbal head in the structure (in this case the complex Aux head).

(63) *Aydâ be pesar-esh ye ketâb* ***hedye*** *na-**dâd-e*** *ast.*
Ayda to son-her a book gift GIVE-PTCP NEG-AUX.PRF.3SG

“Ayda has not gifted a book to her son.”

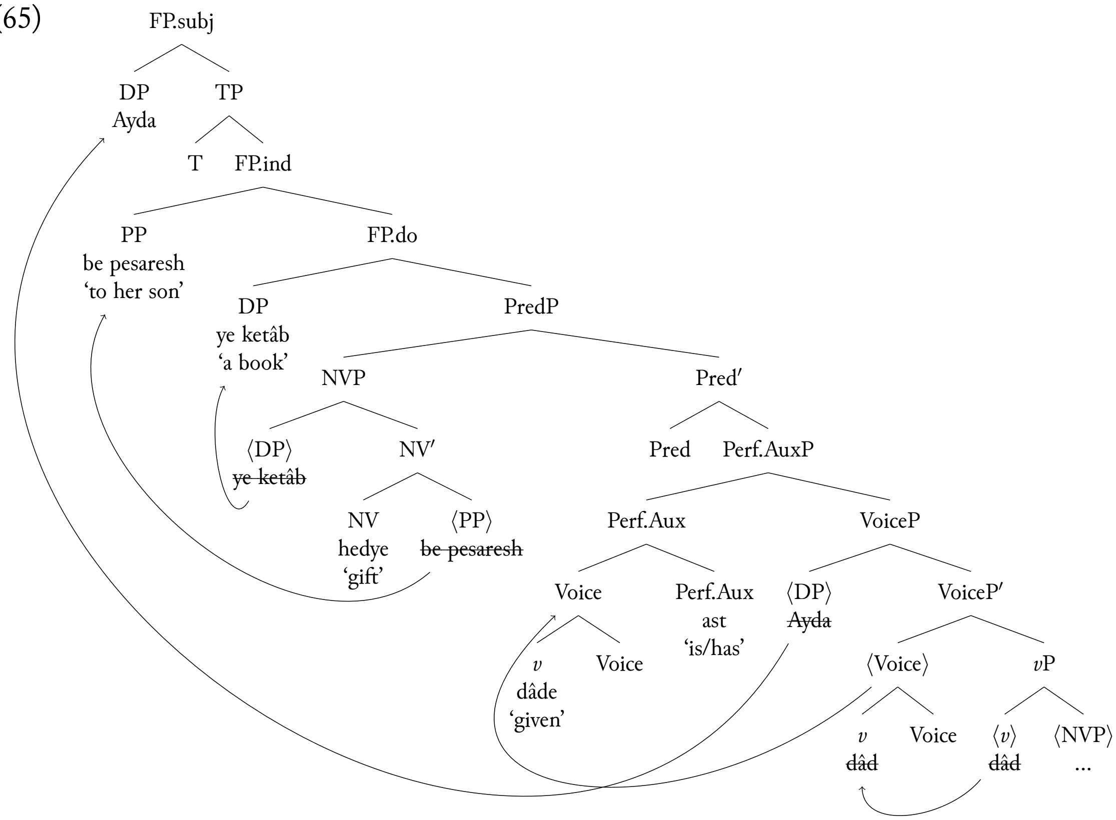
Building on the complex predicate derivation, I assume that the Aux hierarchically projects higher and selects for the VoiceP; this accounts for the subject agreement morphology realized on the Aux as the highest verbal head in (62). The agreement would otherwise show up on the Voice head if it were the highest verb head below PredP, as was the case in (49). Therefore, all that is required to account for (62) is to include an AuxP projection above the *v*P and below the PredP.

(64)



The rest of the derivation proceeds in (65) as proposed above; the NVP preposes to [Spec.PredP], the internal arguments raise to FP.do and FP.ind, and lastly the external argument is licensed in FP.subj.

(65)



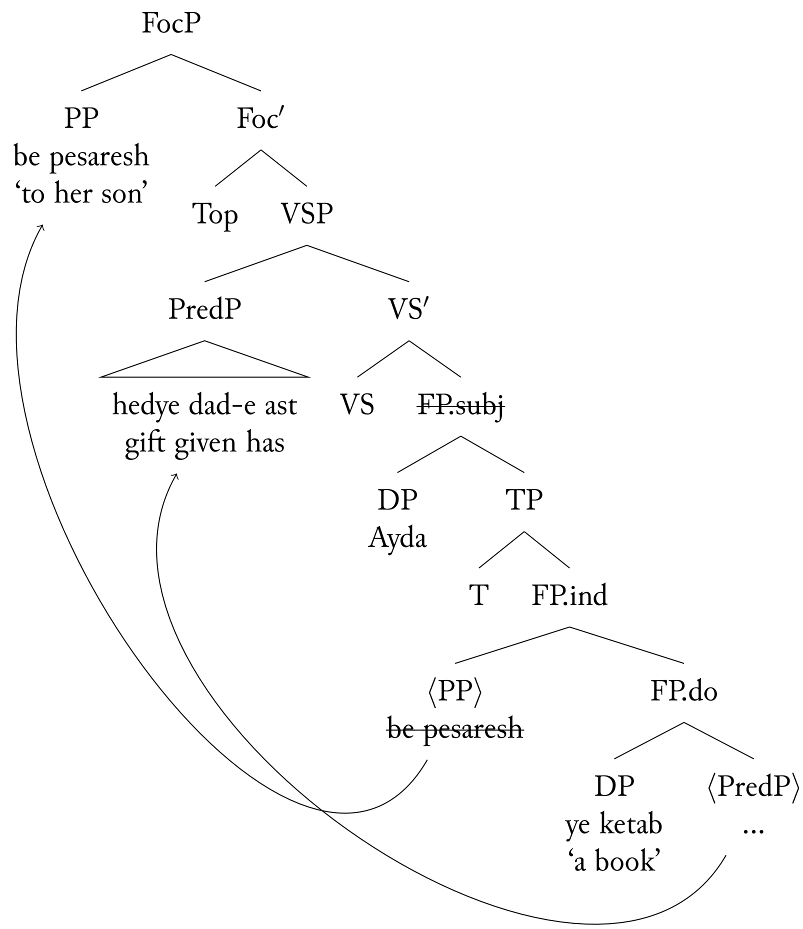
In addition to straightforwardly deriving the canonical Perf.Aux-NV-LV order through uniform leftward movements and left-adjunction only, the derivation in (65) is a desirable outcome for the ellipsis cases in (44), repeated below as (66).

(66) a. *Aydâ be doxtar-esh ye ketâb* ***hedye na-dâd-e*** *ast.*
Ayda to daughter-her a book gift NEG-GIVE-PTCP AUX.PRF.3SG
"Ayda has not gifted a book to her daughter."

b. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~* ~~***hedye***~~ ***dâd-e*** *ast.*
but ~~Ayda~~ to son-her ~~a book gift~~ GIVE-PTCP AUX.PRF.3SG
"But (she) has gifted (a book) to her son."

c. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~* ***hedye dâd-e*** *ast.*
but ~~Ayda~~ to son-her ~~a book~~ gift GIVE-PTCP AUX.PRF.3SG

In (66), the indirect object *be pesaresh* 'to her son' contrasts with *be doxtaresh* 'to her daughter' and survives ellipsis. The verbal clusters *hedye na-dâde ast* 'has not gifted' (in the antecedent clause) and *hedye dâde ast* 'has gifted' (in the ellipsis clause) are also contrastive, with the option of either the Aux+LV in (66b) or the Aux+NV+LV in (66c) escaping ellipsis. The two choices are captured through raising either the Perf.AuxP or the PredP out of the ellipsis site and to the specifier of the Verb Stranding Phrase proposed above. The derivation for the latter option in (66c) is given in (67).

(67) 

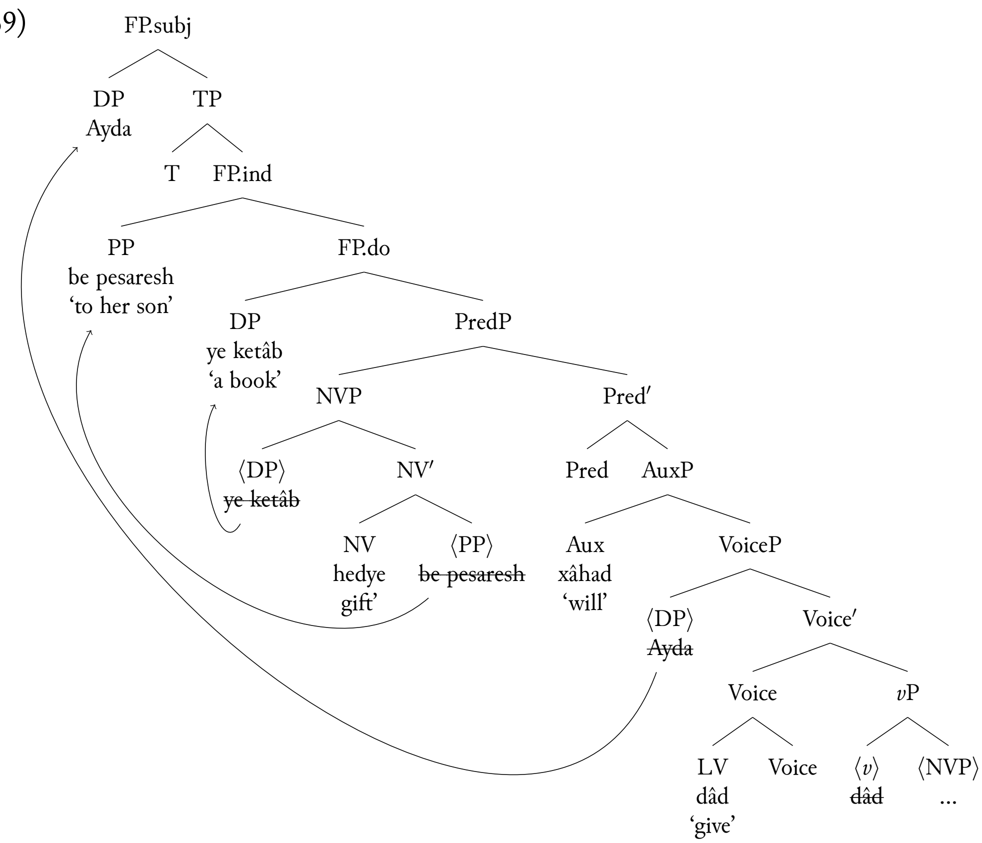
3.3 Complex predicates with the Future Auxiliary

This section offers an analysis for deriving the fact that the Future Aux *xâstan* surfaces penultimately and breaks up the components of a complex predicate in data such as (68).

(68) *Aydâ be pesar-esh ye ketâb* ***hedye*** *xâhad* ***dâd.***
Ayda to son-her a book gift FUT.3SG give
"Ayda will gift a book to her son."

I propose that this auxiliary also hierarchically projects above and selects for the VoiceP to which the light verb adjoins. This is once again supported by the subject agreement inflection on the auxiliary as the highest verbal head in the structure. In comparison to the Perfect Aux, however, the feature checking between the Future Aux and the (short)-infinitival verb it selects for does not involve adjunction to the Aux head before spell-out. Therefore, the inflected aux *xâhad* linearly precedes the LV *dâd* in (68) as a reflection of the hierarchical base order of the two heads. Following previous assumptions, a PredP in turn merges above the Aux projection and preposes the NV projection to its specifier. The rest of the derivation follows as expected in (69); the internal arguments raise to FP.do and FP.ind, and the external argument is licensed in FP.subj.

(69)



This account has the advantage of eliminating the need to assume rightward movement/right-adjunction and the need to commit to an excorporation operation for deriving the strict NV-Fut.Aux-LV order. The derivation of the ellipsis constructions in (31), repeated below as (70), naturally proceeds from the structure in (69).

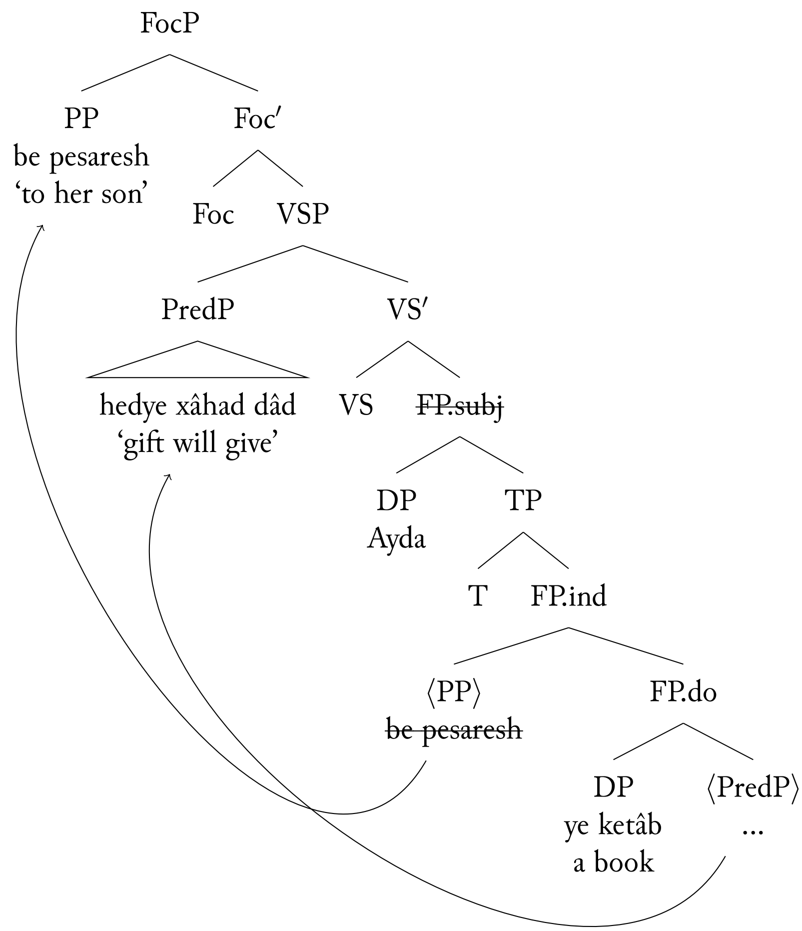
(70) a. *Aydâ be dokhtar-esh ye ketâb* ***hedye*** *na-xâhad* ***dâd***.
Ayda to daughter-her a book gift NEG-FUT.3SG give
"Ayda will not gift a book to her daughter."

b. *Vali ~~Aydâ~~ be pesar-esh ~~ye ketâb~~* ~~***hedye***~~ *xâhad* ***dâd***.
but ~~Ayda~~ to son-her ~~a book gift~~ FUT.3SG give
"But (she) will gift (a book) to her son."

c. *Vali ~~Ayda~~ be pesar-esh ~~ye ketâb~~* ***hedye*** *xâhad* ***dâd***.
but ~~Ayda~~ to son-her ~~a book~~ gift FUT.3SG give

Ultimately, the option for the entire complex cluster *hedye xâhad dâd* 'gift will give' to escape ellipsis in (70c) is captured through the movement of PredP out of the ellipsis site, as shown in (71).

(71)



4 Complex predicate derivation and Topic/Focus fronting

Given the proposed phrasal movement account of verb stranding in section 3 and the already existing independent evidence for leftward scrambling of XPs in Farsi, I further analyze "verb preposing" as an XP movement that targets the highest extended projection of the verb. This approach is in line with a unified treatment of leftward movement in Farsi as broadly XP movement (Ganjavi 2007, Ghomeshi 1996, Kahnemuyipour 2004, 2009; a.o) based primarily on similar data as those discussed in section 2.2.1 of this paper.

Ghomeshi (1996) proposes that while the data documented in Karimi (1989, 1994) suggests for a separation between XP movement as the source of argument scrambling vs. X^0 raising as the source of verb movement, the inseparability of the non-verbal and the verbal components of a complex predicate is not expected under the X^0 movement story. This was observed for *gharz dâdan* 'rent give' in (42), repeated here in (72).

(72) a. *Kimeâ ketâb-ro be dust-esh* ***gharz dâd***.
Kimea book-DOM to friend-her loan gave.3SG
"Kimea loaned the book to her friend."

b. * *KIMEA*$_i$ ***dâd***$_j$ ⟨*Kimeâ*⟩$_i$ *ketâb-ro be dust-esh* ***gharz*** ⟨***dâd***⟩$_j$
Kimea gave.3SG book-DOM to friend-her loan
"It was Kimea who loaned the book to her friend."

c. *KIMEA$_i$* ***gharz dâd$_j$*** ⟨*Kimeâ*⟩$_i$ *ketâb-ro be dust-esh* ⟨***gharz dâd***⟩$_j$
Kimea loan gave.3SG book-DOM to friend-her

Based on Ghomeshi's suggestion that subjects merge extrernal to VP and "definite direct objects and PPs are adjoined to VP" (or alternatively definite/specific DPs raise outside of VP per Karimi (2005)), cases such as (72c) can be analyzed as the leftward movement of the verbal phrase constituent containing just the verb and it's non-verbal XP complement. This also explains Karimi's (1989, 1994) general observation that arguments may only surface in a position following the verb when they are specific/definite and may not do so otherwise. In (73b), the specific direct object *hayât-o* 'the courtyard' can surface after the complex verb *tamiz kardan* 'clean', whereas (74b) shows that the non-specific object *hayât* 'courtyard' may not do so.

(73) a. *Sepide emruz hayât-o* ***tamiz kard.***
Sepide today courtyard-DOM clean did.3SG
"Sepide cleaned the courtyard today."

b. *Sepide emruz* ***tamiz kard*** *hayât-o.*
Sepide today clean did.3SG courtyard-DOM
"Sepide CLEANED the courtyard today" (Karimi 1989: p. 213)

(74) a. *Sepide emruz hayât* ***tamiz kard.***
Sepide today courtyard clean did.3SG
"Sepide cleaned coutyards today."

b. * *Sepide emruz* ***tamiz kard*** *hayât.*
Sepide today clean did.3SG courtyard
(Karimi 1989: p. 213)

Ghomeshi (1996) suggests that this difference can be explained if "XP-preposing", as opposed to object postposing is at play. In (73a), the VP contains only the complex predicate and excludes the specific direct object 'the courtyard' because such objects are external to the VP under either the base-generation or the movement analysis. So, VP should be able to move to a (sentence internal) position above it in (73b). Since non-specific objects are VP internal, however, the object 'yard' in (74a) must move along with the complex predicate whenever VP undergoes leftward movement; so, a VP movement that leaves a non-specific direct object stranded in (74b) is not predicted to be possible.

Ganjavi (2007) discusses the option of forming yes-no questions in Farsi either with the verb in-situ and with rising intonation at the end of the sentence in (75a), or through the movement of the verb to the front in (75c). Ganjavi also observes that such movement targets both the verb and the direct object whenever the object is non-specific. The same holds in (76b) as the modified direct object *film-e xeyli ahmaghane* 'very stupid movie' and the verb 'see' must both front.

(75) a. *Nâder* <u>*ghazâ*</u> ***poxt?***
Nader food cooked.3SG
"Did Nader cook food?"

b. * ***poxt*** *Nâder* <u>*ghazâ*</u> ⟨***poxt***⟩*?*
cooked.3SG Nader food

"Did Nader cook food?"

c. *ghazâ* ***poxt*** *Nâder* ⟨*ghazâ* ***poxt***⟩*?*
food cooked.3SG Nader

(Ganjavi 2007: p. 32)

(76) a. * ***mi-bin-e*** *Rezâ har shab film-e xeyli ahmaghâne* ⟨***mi-bin-e***⟩*?*
DUR-see.3SG Reza every night movie-Ezf very stupid
"Does Reza watch very stupid movies every night?"

b. *film-e xeyli ahmaghâne* ***mi-bin-e*** *Rezâ har shab*
movie-Ezf very stupid DUR-see.3SG Reza every night
⟨*film-e xeyli ahmaghâne* ***mi-bin-e***⟩*?*

(Ganjavi 2007: p. 32)

In line with Ghomeshi (1996), Ganjavi (2007) takes (75c) and (76b) to argue for a "VP fronting" analysis. Ganjavi rules out such cases as involving subject postposing based on the following in (77) and (78). In (77a), the adverb *hanuz* 'still' follows the subject 'Nader' and precedes the objects *bara dustamun* 'for our friends' and *ghazâ* 'food'. As (77c) reveals, the relative ordering of the subject and the adverb remains unchanged when the object and the verb move. As (78) also shows, the preposing of the non-specific direct object *ketab* 'book' and the verb *xaridan* 'to buy' in (78c) does not alter the order in which the subject 'Homeyra' and the adverb *hamishe* 'always' appear. This is to show that the subject hasn't postposed and what's at play is really XP fronting.

(77) a. *Nâder hanuz barâ dust-â-mun ghazâ* ***na-poxt-e****.*
Nader still for frined-PL-our food NEG-cooked-PTCP
"Nader still hasn't cooked food for our friends."

b. * ***na-poxt-e*** *Nâder hanuz barâ dust-â-tun ghazâ* ⟨***na-poxt-e***⟩*?*
NEG-cooked-PTCP Nader still for frined-PL-your food
"Hasn't Nader cooked for your friends yet?"

c. *ghazâ* ***na-poxt-e*** *Nâder hanuz barâ dust-â-tun* ⟨*ghazâ* ***na-poxt-e***⟩*?*
food NEG-cooked-PTCP Nader still for frined-PL-your

(Ganjavi 2007: pp. 34–35)

(78) a. *Homeyrâ hamishe ketâb* ***mi-xa-re****.*
Homeyra always book DUR.buys.3SG
"Homeyra always buys books."

b. * ***mi-xa-re*** *Homeyrâ hamishe ketâb* ⟨***mi-xa-re***⟩*?*
DUR.buys.3SG Homeyra always book
"Does Homeyra always buy books?"

c. *ketâb* ***mi-xa-re*** *Homeyrâ hamishe* ⟨*ketâb* ***mi-xa-re***⟩*?*
book DUR.buys.3SG Homeyra always

(Ganjavi 2007: p. 35)

.

Lastly, Kahnemuyipour (2004, 2009) proposes that the obligatory movement of manner adverbs along with the verbs, as shown in (41b) and further exemplified in (79) and (80), operates on the maximal projections (vP) that contain the adverbs and cannot strand them.

(79) a. *Ali <u>ârum</u> <u>ghazâ</u> mi-xor-e.*
Ali slowly food DUR-eat.3SG
"Ali eats slowly."

b. * *<u>ghazâ</u> mi-xor-e Ali <u>ârum</u> ⟨<u>ghazâ</u> mi-xor-e⟩.*
food DUR-eat.3SG Ali slowly

c. *<u>ârum</u> <u>ghazâ</u> mi-xor-e Ali ⟨<u>ârum</u> <u>ghazâ</u> mi-xor-e⟩.*
slowly food DUR-eat.3SG Ali

(Kahnemuyipour 2009: p. 89)

(80) a. *Ali <u>xub</u>* ***bâzi kard.***
Ali well play did.3SG
"Ali played well."

b. * ***bâzi kard*** *Ali <u>xub</u>* ⟨***bâzi kard***⟩.
play did.3SG Ali well

c. *<u>xub</u>* ***bâzi kard*** *Ali* ⟨*<u>xub</u>* ***bâzi kard***⟩.
well play did.3SG Ali

(Kahnemuyipour 2009: p. 90)

In line with the XP movement view of "verb preposing", I propose that the same Topic/Focus projections to which arguments scramble can host verbal constituents as well.[9] Where fronting

[9]An anonymous reviewer raises the question of whether it is possible to have long distance focus movement of an embedded constituent, accompanied by long distance movement of the verb. If so, an argument can be made that what crosses the clause boundry is a maximal projection and not a head. Karimi (2005) argues that verb fronting can only happen within the same clause and cannot be long distance, as shown in (1) below.

(1) a. *[un film-ro did-an] unâ.*
that movie-râ saw-3PL they
"They saw that movie."

b. * *[un film-ro did-an] Kimea goft unâ.*
that movie-râ saw-3PL Kimea said-3PL they

Karimi analyzes (1a) as the topic movement of the direct object *un film-ro* 'that movie' to [Spec, TP] and the movement of the verb *didan* 'to see' to T. The ungrammaticaly of (1b) is therefore argued to be due to the locality of V movement. However, counter cases of the long-distance movement of the verb can be found, such as in the following from Bonami & Samvelian (2015: ex. 22b).

(2) ***dâd*** *Maryam fekr mi-kon-am [in tâblo-râ be Omid diruz].*
gave.3SG Maryam thought DUR-do.1SG this painting-râ to Omid yesterday
"I think that Maryam gave this painting to Omid yesterday."

In (2), the inflected embedded verb *dâd* 'gave' and the embedded subject 'Maryam' have moved to the matrix clause, and they must have done so separately given that the subject and the verb alone do not form a constituent. The same is possible with the complex predicate *gharz dâd* 'lent' in (3).

(3) ***gharz dâd*** *Maryam fekr mi-kon-am [in tâblo-râ be Omid diruz].*
loan gave.3SG Maryam thought DUR-do.1SG this painting-râ to Omid yesterday
"I think that Maryam lent this painting to Omid yesterday."

seems to diverge from ellipsis to some extent is the strong requirement for Farsi speakers that all verbal elements plus additional material in the extended functional projection of the verb move to the left. This is a restriction that must be encoded in the grammar regardless of whether one assumes the base structure of the VP/*v*P to be head-final or head-initial.l[10] This means that scrambling of the maximal functional projection is what derives the verb preposing facts in cases such as the following in (81). The grammatical case of verb fronting in (81f) shows that the manner adverb *bezur* 'reluctantly'/'forcibly' and the non-specific direct object *ketâb* 'book' must front together with the auxiliary containing cluster *gharz xâhad dâd* 'will loan'.

(81) a. *Kimea be dust-esh be-zur ketâb **gharz** xâhad **dâd**.*
Kimea to friend-her with-force book loan AUX.FUT.3SG give
"Kimea will reluctantly/forcibly lend books to her friend."

b. * ***dâd***$_i$ *Kimeâ be dust-esh be-zur ketâb **gharz** xâhad* ⟨***dâd***⟩$_i$
give Kimea to friend-her with-force book loan AUX.FUT.3sg

c. * *xâhad **dâd***$_i$ *Kimeâ be dust-esh be-zur ketâb **gharz*** ⟨*xâhad **dâd***⟩$_i$
AUX.FUT.3SG give Kimea to friend-her with-force book loan

d. * ***gharz** xâhad **dâd***$_i$ *Kimeâ be dust-esh be-zur ketâb* ⟨***gharz** xâhad **dâd***⟩$_i$
loan FUT.3SG give Kimea to friend-her with-force book

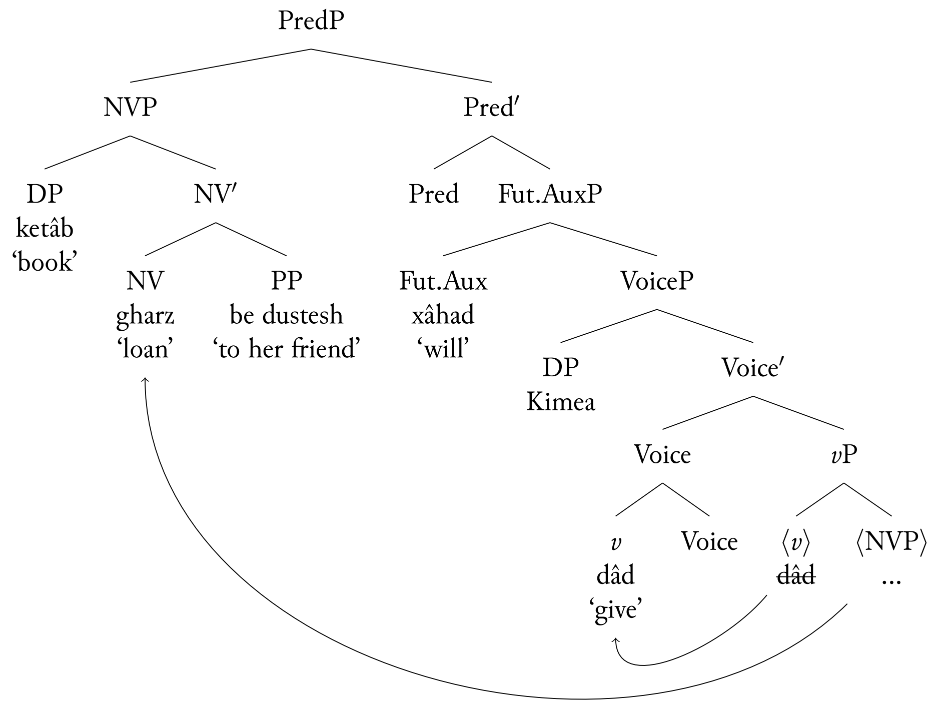
e. * *ketâb **gharz** xâhad **dâd***$_i$ *Kimeâ be dust-esh be-zur*
book loan FUT.3SG give Kimea to friend-her with-force
⟨*ketâb **gharz** xâhad **dâd***⟩$_i$

f. *be-zur ketâb **gharz** xâhad **dâd***$_i$ *Kimeâ be dust-esh*
with-force book loan FUT.3SG give Kimea to friend-her
⟨*be-zur ketâb **gharz** xâhad **dâd***⟩$_i$

To account for the gramamtical movement possibility in (81f), I will first derive the neutral case in (81a). I start with the previously laid out assumptions; (a) the NV predicate (*gharz* 'loan') and its arguments (*be dustesh* 'to her friend' and *book* 'book') form their own NVP, which is further elected for by the light verbal *v* head *dâd* 'give'; (b) the external argument 'Kimea' is introduced externally in VoiceP; the auxiliary (Future Aux *xâstan*) selects for VoiceP; and the PredP merges above the AuxP and licenses the NV predicate locally in its specifier.

(82)

[10] I do not have a conclusive hypothesis as to what this requirement stems from. As far as the Farsi data is concerned, I've shown in relevant examples that "verb movement" necessarily involves the movement of the non-verbal predicate, the light verb, and the auxiliary + certain other phrasal elements (non-specific objects, manner adverbs, etc.). That's why I've analyzed these cases as the topicalization/focalization of the functional projection that will contain all the items that have not moved out or have not originated external to this constituent. To this effect, "verb movement" as discussed here is to be thought of as FP-fronting.

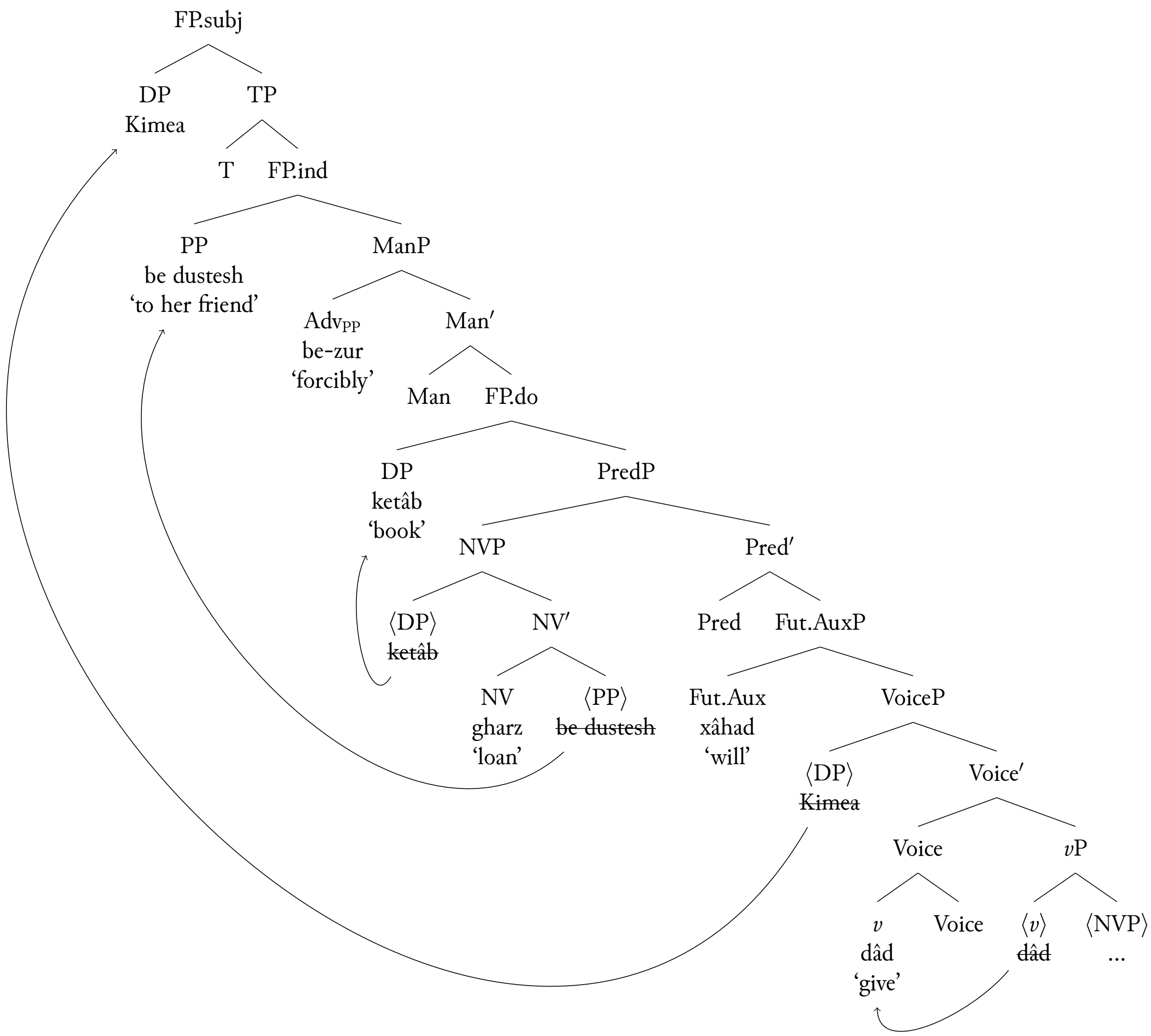


In Kahnemuyipour (2009: p. 89), vP is argued to be the maximal projection that moves in "VP-fronting". Since manner adverbs must front with the verb and cannot be stranded (as evident in (81e)), Kahnemuyipour proposes that these adverbs orignate in a ManP projection below vP.

(83) Persian: . . . $[_{vP}$ t_{subj} v $[_{ManP}$ Adv_{Man} $[_{AspP}$ DP_{obj} Asp $[_{VP}$ V $t_{obj}]$

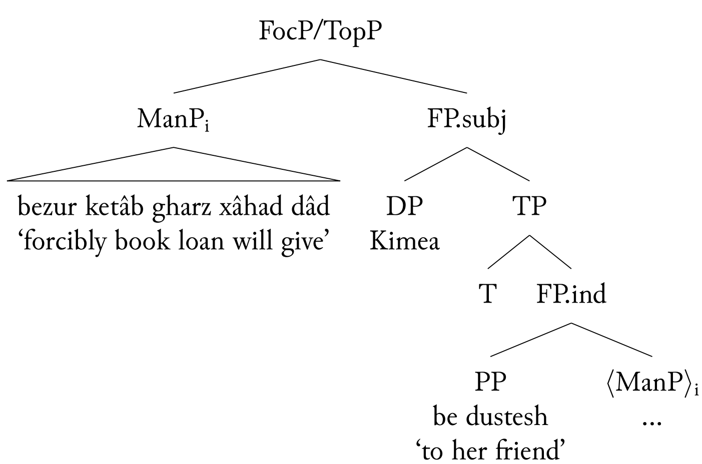
In my analysis however the ManP merges above the Functional Projection to which the direct object always shifts to and below the derived position of the indirect object PP. The rest of the derivation proceeds and the subject is subsequently licensed in the FP.subj.

(84)



The next step of deriving (81f) therefore involves the movement of the ManP to a left-peripheral Focus/Topic position, as suggested in (85):

(85)



5 Conclusion

In this paper, I've argued that the widely adopted VP/vP head final analyses for Farsi run into some theoretical inconsistencies. These inconsistencies were highlighted in the discussion of Rasekhi's (2018) verb-stranding *v*P ellipsis and Karimi's (2005) optional verb fronting/scrambling in sentences involving Farsi complex predicates and the Future auxiliary verb in particular. I've argued that relying on verb movement alone as the source behind ellipsis and fronting runs into inconsistency problems, such as the need to (a) adopt both leftward and rightward movement and adjunction to both the left and the right of licensing heads, (b) resort to head-excorporation of the sort that isn't permitted by the head-adjunction theory of excorporaion in the first place, and (c) posit rules for adjoining non-heads to heads. I've proposed that all phrases in Farsi start from a base Spec-Head-Comp (S-H-C) (Kayne 1994) configuration and that the mixed word orders are derived from leftward syntactic movements that also confirm to S-H-C. By deriving complex predicates through consistent licensing of phrases in designated functional projections, I've shown that verbal elements become clusters contained under maximal XP projections through the course of the derivation. As a result, both ellipsis and verb preposing are analyzed as instances of XP movements to Focus and Topic positions.

Acknowledgements

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Abbreviations

1 = 1st Person, 2 = 2nd person, 3 = 3rd person, AUX.FUT = future auxiliary, AUX.PERF = perfect auxiliary, DOM = differential object marking, DUR = durative, Ezf = ezafe, NEG = negation, PTCP = participle, PL = plural, REL = relative, SG = singular, SBJV = subjunctive

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